

PICARD GROUPS OF QUOTIENT RING SPECTRA: REAL K -THEORY AND LIFTING OBSTRUCTIONS

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ABSTRACT. We compute Picard groups for the descent-tower quotients KO/η^k , their ordinary two-completions, their $K(1)$ -local module categories, and specified $\mathbb{E}_2$ double quotients $KO/(2^m, \eta^n)$ for $m \geq 5$. Ordinary completion preserves the Picard group at every eta stage, whereas local invertibility introduces two-adic families. The calculations use antipodal spherical cochains and the distinction between an invertible local system and one realized by an invertible cochain module. A coherent Bockstein and a connection argument determine the exponent-32 input; a finite Postnikov comparison determines the finite-coefficient extensions. We also construct a rational quotient tower with a non-liftable Picard class at any prescribed stage, and, for every prime $p \geq 5$, a height-two generalized Moore algebra whose $K(2)$ -localization has a Picard class of order p modulo classes coming from local spectra.

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1. INTRODUCTION

A map of $\mathbb{E}_2$ -algebras induces a homomorphism on Picard groups by extension of scalars. For quotient ring spectra, this homomorphism measures which invertible modules lift through a quotient and which arise only after taking it. Even when the underlying quotient is a short cofiber, the multiplication is part of the quotient data, and invertibility can change when the module category is localized. We study these questions through explicit calculations in real K -theory and constructions of non-liftable invertible modules.

The calculations of Levy–Li–Zhang provide the starting point. Their descent tower gives compatible quotient multiplications [LLZ26, Definition 2.10]. They compute the first eta-power stages [LLZ26, Propositions 3.13 and 3.16] and real K -theory modulo 2^m for $m \geq 6$ [LLZ26, Theorem 3.23]. We determine the entire eta-power tower, compare ordinary completion with local invertibility, and extend the finite double-quotient calculation to every eta stage and $m \geq 5$. The exponent-32 calculation and the uniform relative arguments are proved here. The final two constructions address the lifting questions in [LLZ26, Questions 1.5, 1.6, and 3.21].

1.1. The eta-power tower. Throughout, KO and KU are integral periodic real and complex K -theory, and $\eta \in \pi_1 KO$ is the image of the stable Hopf element. All limits, tensor products, and cofibers are derived. Define commutative KO -algebras by

$$(1.1) \quad A_1 = KU, \quad A_{k+1} = A_k \times_{A_k \otimes_{KO} KU} KU \quad (k \geq 1).$$

Both maps into the middle algebra are structural maps; $A_{k+1} \rightarrow A_k$ is the first projection. This is the commutative refinement of the descent tower specified in [LLZ26, Definition 3.3(1)]. The general tower in [LLZ26, Definition 2.10] is a tower of $\mathbb{E}_1$ -algebras; its stage D_{k-1} is our A_k . The Wood sequence and the tower construction give

$$A_k \simeq \text{cofib}(\eta^k : \Sigma^k KO \rightarrow KO) \quad \text{in } \text{Mod}_{KO}.$$

The notation KO/η^k below always means the multiplication in (1.1).

Write $\hat{A}_k = KO_2^\wedge \otimes_{KO} A_k$. For a commutative ring spectrum B , let $\text{Pic}(B)$ denote the Picard group of its ordinary module category. If B is $K(1)$ -local at two, put

$$\text{Pic}_{K(1)}(B) = \text{Pic}(\text{Mod}_B^{K(1)}), \quad M \otimes_B^{K(1)} N = L_{K(1)}(M \otimes_B N).$$

These are different definitions, even when the coefficient ring is already complete. In the following table, $\mathbf{Z}_2$ is the additive group of two-adic integers.

TABLE 1. Ordinary and local Picard groups of the eta-power tower.

Stage	$\text{Pic}(A_k) = \text{Pic}(\hat{A}_k)$	$\text{Pic}_{K(1)}(\hat{A}_k)$
$k = 1$	$\mathbf{Z}/2$	$\mathbf{Z}/2$
$k = 2$	$\mathbf{Z}/4$	$\mathbf{Z}/4 \oplus \mathbf{Z}_2$
$k = 3$	$\mathbf{Z}/4$	$\mathbf{Z}/4$
$k = 4, 5$	$\mathbf{Z}/8$	$\mathbf{Z}/8$
$k \geq 6, k \equiv 2 \pmod{4}$	$\mathbf{Z}/8$	$\mathbf{Z}/8 \oplus \mathbf{Z}_2$
$k \geq 12, k \equiv 4, 5 \pmod{8}$	$\mathbf{Z}/8 \oplus \mathbf{Z}/2$	$\mathbf{Z}/8 \oplus \mathbf{Z}/2$
All remaining $k \geq 6$	$\mathbf{Z}/8$	$\mathbf{Z}/8$

Theorem 1.1 (Ordinary Picard groups and completion). *For every $k \geq 1$, the canonical homomorphism*

$$\text{Pic}(A_k) \longrightarrow \text{Pic}(\hat{A}_k)$$

is an isomorphism, natural in the tower, and these groups are given by the middle column of Table 1. For $k \geq 4$, suspension generates the displayed $\mathbf{Z}/8$ -summand. Its complementary subgroup is canonically the kernel of restriction to stage 4, both before and after completion.

Theorem 1.2 (Local Picard groups). *For every $k \geq 1$, the group $\text{Pic}_{K(1)}(\hat{A}_k)$ is given by the last column of Table 1. For $k \geq 5$, suspension generates the displayed $\mathbf{Z}/8$ -summand, and the complementary subgroup is canonically the kernel of restriction to stage 5.*

The extra ordinary classes first occur at stages 12 and 13. At stages 4 and 5, the same relative classes are present, but a nonzero boundary prevents them from defining extra absolute classes. In the local category, the additional two-adic groups arise from a different condition on spherical local systems: their clutching coefficients need only be nilpotent modulo two.

1.2. Finite double quotients. For $m \geq 6$, choose the compatible Moore $\mathbb{E}_2$ multiplications used in [LLZ26, Sections 2.3 and 3.3]. At $m = 5$, choose an $\mathbb{E}_2$ multiplication obtained by realizing the $\mathbf{F}_2$ -synthetic Moore quotient of [Bur22, Theorem 3.2]; the structure and the needed Bockstein are specified in the exponent-32 proof below. Put

$$KO/(2^m, \eta^n) := A_n \otimes (\mathbb{S}/2^m), \quad m \geq 5, \quad n \geq 1.$$

This fixes the meaning of $\mathrm{KO}/(2^m, \eta^n)$. An $\mathbb{E}_2$ structure gives a monoidal category of modules, hence a group of invertible objects. The abelianness and splittings in the next theorem are part of the calculation. No $\mathbb{E}_3$ refinement or comparison between different choices of m is assumed.

Theorem 1.3 (Finite double quotients). *For the specified multiplications and every $m \geq 5$, the groups $\mathrm{Pic}(\mathrm{KO}/(2^m, \eta^n))$ in the first five stages are*

n	1	2	3	4	5
Picard group	$\mathbf{Z}/2$	$\mathbf{Z}/4 \oplus \mathbf{Z}/2^{m-2}$	$\mathbf{Z}/4 \oplus \mathbf{Z}/2$	$\mathbf{Z}/8 \oplus \mathbf{Z}/2$	$\mathbf{Z}/8$.

Let c denote complexification. For every $n \geq 6$, there is a split decomposition

$$(1.2) \quad \mathrm{Pic}(\mathrm{KO}/(2^m, \eta^n)) \cong \mathbf{Z}/8 \oplus \ker \left(\pi_{-n-2}(\mathrm{KO}/2^m) \xrightarrow{c} \pi_{-n-2}(\mathrm{KU}/2^m) \longrightarrow \pi_{-n-2}(\mathrm{KU}/2^m)/2 \right).$$

The final slash denotes reduction of an abelian group modulo two. The kernel has the following values:

$n \bmod 8$	0	1	2	3	4	5	6	7
kernel	0	0	$\mathbf{Z}/2^m$	$\mathbf{Z}/2$	$(\mathbf{Z}/2)^2$	$\mathbf{Z}/2$	$\mathbf{Z}/2^{m-1}$	0.

The $\mathbf{Z}/8$ -summand is generated by suspension. Its complement is canonically the kernel of restriction

$$\mathrm{Pic}(\mathrm{KO}/(2^m, \eta^n)) \longrightarrow \mathrm{Pic}(\mathrm{KO}/(2^m, \eta^5)).$$

Every module over $\mathrm{KO}/(2^m, \eta^n)$ is already $K(1)$ -local, so the ordinary and local module categories give the same Picard group.

The bound $m \geq 5$ is a bound for these chosen structures. It does not assign a Picard group to an unspecified multiplication on a double cofiber. The low-exponent results in the appendix give separate nonexistence statements for specified units and record the scope of intrinsic complex quotient structures.

1.3. Lifting obstructions. The eta calculation itself gives non-surjective transition maps: for every $r \geq 1$,

$$\mathrm{coker}(\mathrm{Pic}(A_{8r+6}) \longrightarrow \mathrm{Pic}(A_{8r+5})) \cong \mathbf{Z}/2.$$

The following rational construction allows the failing transition to be prescribed.

Theorem 1.4 (A rational obstruction at a prescribed stage). *For every $m \geq 1$, there are a rational $\mathbb{E}_\infty$ -ring R_m and $t \in \pi_0 R_m$, with compatible $\mathbb{E}_\infty$ quotients R_m/t^j , such that*

$$\mathrm{Pic}(R_m/t^{m+1}) \longrightarrow \mathrm{Pic}(R_m/t^m)$$

is injective but not surjective. Its cokernel contains an element of infinite order, and $\pi_{-1}(R_m/t) = 0$. The underlying R_m -module of R_m/t^j is the cofiber of multiplication by t^j .

The ring R_m is allowed to depend on m . It is the derived ring of functions on the complement of the origin in $xy - zw = t^m$. A line bundle on the punctured central quadric extends to the m th neighborhood, but a nonzero Čech obstruction prevents its extension to the next one.

For the last statement, let $\mathrm{Sp}_{K(2)}$ have tensor product $L_{K(2)}(- \wedge -)$, and let BP denote the Brown–Peterson spectrum at the indicated prime. We use Araki generators, with $v_1 \in BP_{2(p-1)}$.

Theorem 1.5 (A height-two obstruction). *For every prime $p \geq 5$, there is a finite generalized Moore $\mathbb{E}_2$ -algebra M of type two with*

$$BP_*M \cong BP_*/(p^4, v_1^{p^4})$$

such that the cokernel of extension of scalars

$$\mathrm{Pic}(\mathrm{Sp}_{K(2)}) \longrightarrow \mathrm{Pic}(L_{K(2)}M)$$

contains a subgroup isomorphic to $\mathbf{Z}/p$. More precisely, there is an invertible module L of order p such that $[L]^r$ is outside the image for every $1 \leq r < p$. Every ordinary module over $L_{K(2)}M$ is already $K(2)$ -local.

Here the obstruction is detected by a crossed homomorphism on the height-two Morava stabilizer group. Its value at a central scalar is nonconstant in a finite Lubin–Tate quotient, whereas every class lifted from the full Lubin–Tate ring has scalar value there. Picard descent realizes this algebraic obstruction by an invertible module.

1.4. The proofs and conventions. The common geometric input identifies the complexification of A_k with cochains on the antipodal sphere $S(k\sigma)$, where σ is the sign representation of C_2 . The cochain comparison then determines which local systems come from modules. Comparing units proves that ordinary completion preserves the Picard group. The Rezk logarithm then identifies the completed relative Picard spectrum with a connective truncation of a shift of $\mathrm{KU}_2^\wedge$, whose homotopy fixed points give a complexification kernel. The boundary prevents the relative classes at stages four and five from lifting to absolute Picard classes; at later stages it vanishes. Restriction to stage four supplies the retraction that splits off the suspension subgroup.

For local modules, the realizability condition changes the lattice in degree zero of this relative spectrum. For finite coefficients, the coefficient algebra is only $\mathbb{E}_2$; the proof instead compares finite Postnikov intervals of Picard spaces. The exponent-32 calculation supplies the low unit differential, and the finite comparison determines the remaining groups and their extensions. The rational and height-two constructions then give two further mechanisms for failure of lifting.

For a commutative ring spectrum B , $\mathrm{pic}(B)$ denotes its connective Picard spectrum, with $\pi_0 \mathrm{pic}(B) = \mathrm{Pic}(B)$. For an $\mathbb{E}_2$ -ring, $\mathcal{P}(B)$ denotes its grouplike $\mathbb{E}_1$ Picard space. Homotopy fixed points of a connective Picard spectrum need not be connective; Picard descent uses its connective cover. We retain the untruncated fixed points whenever a negative homotopy group occurs in a boundary map.

We write $C^*(X; B) = F(\Sigma^\infty X_+, B)$. An action on a spectrum means an object of $\mathrm{Fun}(BC_2, \mathrm{Sp})$; representation suspensions are restricted to this category. The symbols $S(V)$ and S^V denote the unit sphere and one-point compactification of a real representation V . Truncations of spectra with action are objectwise. Numbered references to arXiv papers use the versions specified in the bibliography.

2. SPHERICAL COCHAINS AND THE DESCENT TOWER

For a commutative ring spectrum R , the positive homotopy groups of its Picard spectrum are

$$(2.1) \quad \pi_1 \mathrm{pic}(R) = (\pi_0 R)^\times, \quad \pi_i \mathrm{pic}(R) = \pi_{i-1} R \quad (i \geq 2).$$

Indeed, the loop space at the unit is $\mathrm{GL}_1(R)$; translation from its unit component identifies its positive homotopy groups with those of R [MS16, Section 2.2].

Let g be the nontrivial element of C_2 , acting on KU by conjugation, and write c for complexification. We use the coefficient rings

$$(2.2) \quad \begin{aligned} \pi_* \mathrm{KU} &= \mathbf{Z}[u^{\pm 1}], & |u| &= 2, & g(u) &= -u, \\ \pi_* \mathrm{KO} &= \mathbf{Z}[\eta, \alpha, \beta^{\pm 1}] / (2\eta, \eta^3, \eta\alpha, \alpha^2 - 4\beta), & |\eta| &= 1, & |\alpha| &= 4, & |\beta| &= 8. \end{aligned}$$

Complexification is a faithful C_2 -Galois extension and fits into the Wood cofiber sequence

$$(2.3) \quad \Sigma \mathrm{KO} \xrightarrow{\eta} \mathrm{KO} \longrightarrow \mathrm{KU} \quad \text{in } \mathrm{Mod}(\mathrm{KO}).$$

These inputs are recorded in [Rog08, Section 5.3, Proposition 5.3.1 and equation (5.3.2)]. Completion replaces the free coefficients in (2.2) by $\mathbf{Z}_2$ and preserves the displayed finite two-torsion.

Lemma 2.1. *The groups $\text{Pic}(\text{KU})$ and $\text{Pic}(\text{KU}_2^\wedge)$ are $\mathbf{Z}/2$, generated by suspension. The group $\text{Pic}(\text{KO})$ is $\mathbf{Z}/8$, generated by suspension.*

Proof. The even-periodic regular-coefficient theorem [MS16, Theorem 2.4.6 and Remark 2.4.8] applies to KU and $\text{KU}_2^\wedge$: their degree-zero rings $\mathbf{Z}$ and $\mathbf{Z}_2$ are regular noetherian and have trivial algebraic Picard groups. The resulting graded Picard group records parity, since u is a degree-two unit and there is no odd-degree unit. The assertion for KO is [MS16, Theorem 7.1.2]. $\square$

2.1. The geometric model.

Lemma 2.2. *The structural maps of the tower give equivalences*

$$\text{fib}(\text{KO} \rightarrow A_k) \simeq \Sigma^k \text{KO}, \quad A_k \simeq \text{cofib}(\eta^k : \Sigma^k \text{KO} \rightarrow \text{KO}) \quad \text{in } \text{Mod}(\text{KO}).$$

Moreover $\text{KO}_2^\wedge \otimes_{\text{KO}} \text{KU} \simeq \text{KU}_2^\wedge$, and $\hat{A}_k = \text{KO}_2^\wedge \otimes_{\text{KO}} A_k$ is the two-completion of A_k . Base change identifies its multiplication with the recursively defined tower over $\text{KO}_2^\wedge$ starting at $\text{KU}_2^\wedge$.

Proof. Put $I = \text{fib}(\text{KO} \rightarrow \text{KU})$ and $J_k = \text{fib}(\text{KO} \rightarrow A_k)$. The total fiber of the square

$$\begin{array}{ccc} \text{KO} & \longrightarrow & A_k \\ \downarrow & & \downarrow \\ \text{KU} & \longrightarrow & A_k \otimes_{\text{KO}} \text{KU} \end{array}$$

is $\text{fib}(\text{KO} \rightarrow A_{k+1})$. Computing it horizontally and using exactness of tensor product gives

$$J_{k+1} \simeq \text{fib}(J_k \rightarrow J_k \otimes_{\text{KO}} \text{KU}) \simeq J_k \otimes_{\text{KO}} I.$$

Under this equivalence the map to KO is the composite $J_k \otimes I \rightarrow J_k \rightarrow \text{KO}$. Induction therefore identifies it with the tensor product of $k+1$ copies of $I \rightarrow \text{KO}$. The Wood sequence identifies $I \rightarrow \text{KO}$ with multiplication by η , giving the assertions about the underlying module.

Base-changing the Wood sequence gives a finite cofiber of two-complete spectra. The map from KU to that cofiber is a mod-two equivalence, because the completion map for KO has this property. Thus the cofiber is $\text{KU}_2^\wedge$. The same argument applied to η^k proves that $\hat{A}_k$ is the completion of A_k . Finally, extension of scalars preserves finite limits of commutative algebras: their underlying limits are limits of modules, and extension of scalars on modules is exact. Applying this to each defining pullback identifies the completed tower, including its structural maps. $\square$

Proposition 2.3 (Antipodal sphere model). *There are natural equivalences of commutative algebras with semilinear C_2 -action*

$$(2.4) \quad \begin{aligned} \text{KU} \otimes_{\text{KO}} A_k &\simeq C^*(S(k\sigma); \text{KU}), \\ \text{KU}_2^\wedge \otimes_{\text{KO}_2^\wedge} \hat{A}_k &\simeq C^*(S(k\sigma); \text{KU}_2^\wedge). \end{aligned}$$

On the left, conjugation acts on the displayed complex K -theory factor. On the right, the action combines coefficient conjugation with the antipodal action on the sphere. The maps from stage $k+1$ to stage k are restrictions along the equatorial inclusions $S(k\sigma) \hookrightarrow S((k+1)\sigma)$.

Proof. The Galois equivalence identifies $\text{KU} \otimes_{\text{KO}} \text{KU}$ with $F(C_{2+}, \text{KU})$. Its formula $a \otimes b \mapsto (\gamma \mapsto a\gamma(b))$ carries conjugation on the first tensor factor to

$$(h \cdot f)(\gamma) = h(f(h^{-1}\gamma)), \quad h, \gamma \in C_2.$$

This proves the first-stage assertion, including the action.

For finite CW complexes X, Y , dualizability gives the multiplicative equivalence

$$C^*(X; \mathrm{KU}) \otimes_{\mathrm{KU}} C^*(Y; \mathrm{KU}) \simeq C^*(X \times Y; \mathrm{KU}).$$

Cochains send homotopy colimits of spaces to limits of commutative algebras. Since base change preserves the pullback defining A_{k+1} , the join presentation

$$X * C_2 \simeq \mathrm{hocolim}(X \longleftarrow X \times C_2 \longrightarrow C_2)$$

identifies the complexified k th stage with $C^*(C_2^{*k}; \mathrm{KU})$. The join C_2^{*k} is equivariantly homeomorphic to $S(k\sigma)$: join coordinates with weights t_i and signs ϵ_i map to $(\epsilon_i \sqrt{t_i})_{i=1}^k$. Adding a final factor realizes the equatorial inclusion. This proves naturality along the tower. The completed formula follows by extension of scalars, using finiteness of the sphere and Lemma 2.2. $\square$

Lemma 2.4 (Picard descent). *There are natural equivalences*

$$(2.5) \quad \begin{aligned} \mathrm{pic}(A_k) &\simeq \tau_{\geq 0}(\mathrm{pic}(\mathrm{KU} \otimes_{\mathrm{KO}} A_k)^{hC_2}), \\ \mathrm{pic}(\hat{A}_k) &\simeq \tau_{\geq 0}(\mathrm{pic}(\mathrm{KU}_2^\wedge \otimes_{\mathrm{KO}_2^\wedge} \hat{A}_k)^{hC_2}), \\ \mathrm{pic}(\mathrm{KO}) &\simeq \tau_{\geq 0}(\mathrm{pic}(\mathrm{KU})^{hC_2}). \end{aligned}$$

They commute with the tower maps, completion, and extension of scalars from the respective real K -theory spectra.

Proof. Faithful Galois extensions are preserved by arbitrary commutative base change [Rog08, Lemma 7.1.1]. Apply symmetric monoidal Galois descent [MS16, Theorem 3.3.1] and then the limit-preserving Picard-space functor [MS16, Proposition 2.2.3]. The connective cover occurs because a limit of connective spectra is computed by taking the connective cover of its limit in spectra; see [MS16, equations (3.3)–(3.4)]. Every map used here is structural, which gives the asserted compatibilities. $\square$

2.2. Which local systems come from cochains.

Lemma 2.5 (Cochain comparison). *Let R be a commutative ring spectrum and X a space. Evaluation gives an exact symmetric monoidal, fully faithful functor*

$$(2.6) \quad \mathrm{Perf}(C^*(X; R)) \longrightarrow \mathrm{Fun}(X, \mathrm{Mod}(R)),$$

whose image is the thick subcategory generated by the constant local system R . The induced map

$$(2.7) \quad \mathrm{pic}(C^*(X; R)) \longrightarrow F(\Sigma^\infty X_+, \mathrm{pic}(R))$$

is injective on π_0 and an isomorphism on every π_i , $i \geq 1$. These maps are natural in R and X .

Proof. The family of evaluation maps gives the constant local system R an action of $C^*(X; R)$. Extension of scalars along this family defines (2.6) and is exact and symmetric monoidal. Its map on endomorphisms of the unit is the identity under

$$\mathrm{End}_{C^*(X; R)}(C^*(X; R)) \simeq C^*(X; R) \simeq \mathrm{End}_{\mathrm{Fun}(X, \mathrm{Mod}(R))}(R).$$

Closing first one and then the other variable of the mapping spectrum under finite cofiber sequences and retracts proves full faithfulness on perfect modules. It also identifies the image with the thick subcategory of the constant unit. This is the comparison underlying [MS16, Proposition 4.2.2].

An invertible module is perfect: tensoring with its inverse identifies its mapping functor with that of the compact unit, composed with a colimit-preserving equivalence. The functor therefore restricts to a fully faithful functor on invertible objects. Invertible objects of the target are local systems valued in the Picard groupoid of R . Consequently the map of

Picard spaces includes a union of components and is an equivalence on all based loop spaces. Passing to spectra gives (2.7) and the homotopy-group assertions. Evaluation and extension of scalars also give the claimed naturality. $\square$

Lemma 2.6 (Nilpotence in the thick subcategory). *Let T be an associative ring spectrum and $x \in \pi_d T$ homogeneous. The left T -modules M for which $x^N \pi_* M = 0$ for some $N \geq 1$ form a thick subcategory. If $T \rightarrow R$ is an augmentation and x maps to zero, every object of $\text{Thick}_T(R)$ has this property.*

Proof. The condition is preserved by suspensions and retracts. In a cofiber sequence $L \rightarrow M \rightarrow N$, suppose $x^a \pi_* L = 0$ and $x^b \pi_* N = 0$. For $y \in \pi_* M$, the element $x^b y$ lifts from $\pi_* L$, so $x^{a+b} y = 0$. Rotation of the cofiber sequence proves closure whenever any two terms satisfy the condition. The argument uses the module-linear long exact sequence and does not require x to be central. An augmentation-zero element annihilates $\pi_* R$, proving the last assertion. $\square$

Proposition 2.7 (Spherical realizability). *For $R \in \{\text{KU}, \text{KU}_2^\wedge\}$ and every $n \geq 1$, extension of scalars induces*

$$\text{Pic}(R) \xrightarrow{\sim} \text{Pic}(C^*(S^n; R)) = \mathbf{Z}/2.$$

Thus the invertible local systems realized by perfect cochain modules are exactly the constant suspensions. The displayed isomorphisms commute with completion.

Proof. By Lemmas 2.1 and 2.5, after a constant suspension a realized invertible local system has fiber R and belongs to the thick subcategory of the constant unit. For a connected based space X , transport identifies R -module local systems with modules over $R[\Omega X] = R \otimes \Sigma_+^\infty \Omega X$: the inverse construction is the bar construction, using $B\Omega X \simeq X$. Under this identification the constant unit is the augmentation module.

For $n \geq 2$, a local system with chosen fiber R is classified by

$$\xi \in \pi_n \text{pic}(R) = \pi_{n-1} R.$$

This group vanishes for even n . For odd $n \geq 3$, write $\xi = au^{(n-1)/2}$, where a belongs to $\mathbf{Z}$ or $\mathbf{Z}_2$ according to the choice of R . The adjoint of the identity of S^n gives $S^{n-1} \rightarrow \Omega S^n$. Subtracting the constant map at the identity loop after stabilization gives a class

$$x \in \pi_{n-1}(R[\Omega S^n])$$

whose augmentation is zero. Looping the classifying map shows that x acts on the chosen fiber by multiplication by ξ . If $a \neq 0$, then

$$x^N \cdot 1 = a^N u^{N(n-1)/2} \neq 0 \quad (N \geq 1),$$

since both coefficient rings are integral domains. This contradicts Lemma 2.6. Thus $\xi = 0$.

For $n = 1$, the group algebra is $R[z^{\pm 1}]$ and monodromy on the fiber is a unit $a \in (\pi_0 R)^\times$. The augmentation-zero element $z - 1$ acts by $a - 1$. Lemma 2.6 implies $(a - 1)^N = 0$ for some N , hence $a = 1$. In the completed case the condition is actual nilpotence, not merely two-adic topological nilpotence.

Every realized invertible local system is therefore a constant suspension. Such systems are realized by the corresponding cochain suspensions, and full faithfulness identifies their classes. Since all maps are extension-of-scalars maps, the result commutes with completion. $\square$

2.3. The relative Picard spectrum. For $k \geq 2$, define spectra with C_2 -action by

$$(2.8) \quad \begin{aligned} \text{Rel}_\eta(k) &= \text{cofib}(\text{pic}(\text{KU}) \rightarrow \text{pic}(\text{KU} \otimes_{\text{KO}} A_k)), \\ \widehat{\text{Rel}}_\eta(k) &= \text{cofib}(\text{pic}(\text{KU}_2^\wedge) \rightarrow \text{pic}(\text{KU}_2^\wedge \otimes_{\text{KO}_2^\wedge} \widehat{A}_k)). \end{aligned}$$

Completion defines a map between them, and the tower maps give relative towers before any choice of Bott equivalence.

Proposition 2.8. *For $k \geq 2$ there are equivalences in $\text{Fun}(BC_2, \text{Sp})$*

$$(2.9) \quad \begin{aligned} \text{Rel}_\eta(k) &\simeq \tau_{\geq 1}(\Sigma^{1-k\sigma} \text{pic}(\text{KU})), \\ \widehat{\text{Rel}}_\eta(k) &\simeq \tau_{\geq 1}(\Sigma^{1-k\sigma} \text{pic}(\text{KU}_2^\wedge)). \end{aligned}$$

These equivalences are compatible with completion.

Proof. We give the integral argument; the completed one is identical. Propositions 2.3 and 2.7 show that the map defining $\text{Rel}_\eta(k)$ is an isomorphism on π_0 . Its cofiber is therefore 1-connective. The representation-sphere cofiber sequence

$$S(k\sigma)_+ \longrightarrow S^0 \longrightarrow S^{k\sigma}$$

and Lemma 2.5 give a diagram of cofiber sequences

$$\begin{array}{ccccc} \text{pic}(\text{KU}) & \longrightarrow & \text{pic}(\text{KU} \otimes_{\text{KO}} A_k) & \longrightarrow & \text{Rel}_\eta(k) \\ \parallel & & \downarrow & & \downarrow \\ \text{pic}(\text{KU}) & \longrightarrow & F(\Sigma^\infty S(k\sigma)_+, \text{pic}(\text{KU})) & \longrightarrow & \Sigma^{1-k\sigma} \text{pic}(\text{KU}). \end{array}$$

The middle vertical map is an isomorphism in positive homotopy degrees. After forgetting the action, evaluation at any point splits the lower left map; it is consequently injective on π_0 . The long exact sequences now show that the right vertical map is an isomorphism on every π_i for $i \geq 1$. For $i = 1$ this uses the just-established injectivity and the isomorphism on π_0 in the upper row. The map factors through the 1-connective cover and is an equivalence there. An underlying equivalence is an equivalence of spectra with action. No equivariant point was chosen. Naturality of the diagram gives compatibility with completion. $\square$

3. ORDINARY PICARD GROUPS OF THE ETA-POWER QUOTIENTS

3.1. The first five stages.

Proposition 3.1. *The groups*

$$\text{Pic}(A_1) = \mathbf{Z}/2, \quad \text{Pic}(A_2) = \text{Pic}(A_3) = \mathbf{Z}/4, \quad \text{Pic}(A_4) = \text{Pic}(A_5) = \mathbf{Z}/8$$

are generated by suspension.

Proof. We first obtain finite upper bounds from descent over $\mathbf{RP}^{k-1}$, and then determine the order of suspension. This suffices to determine the groups and their additive extensions.

The cardinality bounds. The cofiber of (2.7) has no positive homotopy groups. Coconnective spectra are closed under limits, so taking homotopy fixed points preserves this property. The long exact sequence and Lemma 2.4 give an injection

$$(3.1) \quad \text{Pic}(A_k) \hookrightarrow \pi_0(F(\Sigma^\infty S(k\sigma)_+, \text{pic}(\text{KU}))^{hC_2}).$$

The target is the group of degree-zero sections of the spectrum local system on $\mathbf{RP}^{k-1}$ associated to the cover $S(k\sigma) \rightarrow \mathbf{RP}^{k-1}$ and the conjugation action on $\text{pic}(\text{KU})$. Indeed, equivariant maps from the sphere give sections over its Borel construction, which is equivalent to its quotient because the action is free.

For a spectrum local system $\mathcal{L}$ on a finite d -dimensional CW complex Y , restriction to the skeleta gives

$$(3.2) \quad E_2^{s,t} = H^s(Y; \pi_t \mathcal{L}) \implies \pi_{t-s} \Gamma(Y; \mathcal{L}), \quad 0 \leq s \leq d.$$

Over an s -cell, relative sections are the s -fold loops of the fiber; the first differential is the cellular coboundary with local coefficients. This constructs the displayed spectral sequence. There are only $d + 1$ skeletal layers, so the filtration is finite, exhaustive, and separated. Thus the order of its degree-zero abutment is at most $\prod_{s=0}^d |H^s(Y; \pi_s \mathcal{L})|$ whenever these groups are finite.

By (2.1) and Lemma 2.1, the first five coefficient groups of $\text{pic}(\text{KU})$ are

$$(\pi_i \text{pic}(\text{KU}))_{i=0}^4 = (\mathbf{Z}/2, \mathbf{Z}/2, 0, \mathbf{Z}^{\text{sign}}, 0),$$

where $\mathbf{Z}^{\text{sign}}$ is the sign module and the first two actions are trivial. For $d = 1, 2$ the nonzero diagonal groups in (3.2) are $H^0(\mathbf{RP}^d; \mathbf{Z}/2)$ and $H^1(\mathbf{RP}^d; \mathbf{Z}/2)$, both $\mathbf{Z}/2$. For $d = 3, 4$ one further group occurs: $H^3(\mathbf{RP}^d; \mathbf{Z}^{\text{sign}}) = \mathbf{Z}/2$. To check the last assertion, the sign-coefficient cellular complex, truncated in degree d , is

$$\mathbf{Z} \xrightarrow{2} \mathbf{Z} \xrightarrow{0} \mathbf{Z} \xrightarrow{2} \mathbf{Z} \xrightarrow{0} \mathbf{Z}.$$

It follows that

$$(3.3) \quad |\text{Pic}(A_2)|, |\text{Pic}(A_3)| \leq 4, \quad |\text{Pic}(A_4)|, |\text{Pic}(A_5)| \leq 8.$$

The order of suspension. For any commutative ring spectrum R , an equivalence $\Sigma^d R \simeq R$ is equivalent to a homogeneous unit in $\pi_d R$: an R -linear map is represented by such a class, and its inverse gives the inverse unit. The image of β makes the suspension class of every A_k have order dividing eight. The long exact sequence of Lemma 2.2 contains

$$\pi_0 \text{KO} = \mathbf{Z} \xrightarrow{\eta^2} \pi_2 \text{KO} = \mathbf{Z}/2 \longrightarrow \pi_2 A_2 \longrightarrow \pi_{-1} \text{KO} = 0.$$

The first map is onto, so $\pi_2 A_2 = 0$ and period one or two is impossible. The bound (3.3) forces suspension to have order four and to generate $\text{Pic}(A_2)$. The map $A_3 \rightarrow A_2$ and the same bound give the assertion for A_3 .

For A_4 , the coefficient vanishing and $\eta^4 = 0$ give $\pi_4 \text{KO} \xrightarrow{\sim} \pi_4 A_4$. The Wood sequence gives

$$0 \longrightarrow \pi_4 \text{KO} \xrightarrow{c} \pi_4 \text{KU} \longrightarrow \pi_2 \text{KO} \longrightarrow 0,$$

hence $c(\alpha) = \pm 2u^2$. Every element of $\pi_4 A_4$ therefore maps to an even multiple of u^2 under $A_4 \rightarrow \text{KU}$. Such an element cannot be a unit. Suspension on A_4 has order eight, and (3.3) identifies the entire Picard group. The map $A_5 \rightarrow A_4$ proves the last case. The first stage is Lemma 2.1. $\square$

3.2. Comparison with two-completion. The proof of completion invariance must account for the enlargement of the unit group. The additive comparison groups are uniquely two-divisible; scalar units supply the remaining lifts.

Lemma 3.2. *The additive group $\mathbf{Z}_2/\mathbf{Z}$ is uniquely two-divisible, and the group $\mathbf{Z}_2^\times/\{\pm 1\}$ has no nonzero two-torsion.*

Proof. For $a \in \mathbf{Z}_2$, subtract an integer with the same parity and divide by two; this proves surjectivity on $\mathbf{Z}_2/\mathbf{Z}$. If $2a \in \mathbf{Z}$, then $2a \in 2\mathbf{Z}_2 \cap \mathbf{Z} = 2\mathbf{Z}$, so $a \in \mathbf{Z}$ and injectivity follows. If a unit class has square one, a representative satisfies $a^2 = 1$ or $a^2 = -1$. The second possibility is excluded modulo eight. The first implies $a = \pm 1$, since $\mathbf{Z}_2$ is an integral domain. $\square$

Lemma 3.3. *Let T be a bounded-below spectrum with C_2 -action whose homotopy groups are uniquely two-divisible. Restriction induces*

$$\pi_j(T^{hC_2}) \xrightarrow{\sim} (\pi_j T)^{C_2} \quad \text{for all } j.$$

In particular, homotopy fixed points preserve its lower connectivity bound.

Proof. Averaging by $(1+g)/2$ makes invariants exact on $\mathbf{Z}[1/2][C_2]$ -modules. Thus the fixed points of each Eilenberg–Mac Lane layer have only the invariant group in the original degree. Induction through a finite Postnikov tower gives the assertion for bounded Postnikov truncations. Now $T \simeq \text{holim}_N \tau_{\leq N} T$, and homotopy fixed points commute with this limit. In each fixed degree and its adjacent degree the computed groups are eventually constant; the Milnor exact sequence has no derived-limit term. This proves the assertion for T . $\square$

Lemma 3.4 (Transfer). *For a spectrum E with an action of a finite group H , there is a natural transfer on underlying spectra $\text{tr} : E \rightarrow E^{hH}$ such that*

$$\text{res tr}(x) = \sum_{h \in H} hx \quad (x \in \pi_j E).$$

Proof. Induction from the trivial subgroup is a finite coproduct of copies of the underlying spectrum of E ; coinduction is the corresponding finite product. They agree in spectra, and the homotopy fixed points of this induced spectrum identify with the underlying E by the coinduction adjunction. Taking fixed points of the induction counit defines the transfer. Restriction is the diagonal into the finite product, and the counit sums the action maps, proving the formula and naturality. $\square$

For $k \geq 2$, define the completion comparison spectrum

$$(3.4) \quad \text{Comp}_\eta(k) = \text{cofib}(\text{pic}(\text{KU} \otimes_{\text{KO}} A_k) \rightarrow \text{pic}(\text{KU}_2^\wedge \otimes_{\text{KO}_2^\wedge} \hat{A}_k)).$$

The reduced completion quotient is the group

$$(3.5) \quad \begin{aligned} \text{RedComp}_\eta(k) &= \text{coker}(\ker[\pi_0 C^*(S(k\sigma); \text{KU}) \rightarrow \pi_0 \text{KU}] \\ &\rightarrow \ker[\pi_0 C^*(S(k\sigma); \text{KU}_2^\wedge) \rightarrow \pi_0 \text{KU}_2^\wedge]). \end{aligned}$$

The maps to the coefficient groups are evaluations. Since $k \geq 2$, the sphere is connected and these maps on π_0 are independent of the chosen point. Their kernels, and hence the quotient, inherit the semilinear C_2 -action of Proposition 2.3.

Lemma 3.5. *The spectrum $\text{Comp}_\eta(k)$ is 1-connective. Its homotopy groups in degrees at least two are finite sums of copies of $\mathbf{Z}_2/\mathbf{Z}$, and, equivariantly,*

$$(3.6) \quad \pi_1 \text{Comp}_\eta(k) \cong \mathbf{Z}_2^\times / \{\pm 1\} \oplus \text{RedComp}_\eta(k).$$

The scalar-unit summand has trivial action. As underlying abelian groups,

$$\text{RedComp}_\eta(k) \cong \begin{cases} 0, & k \text{ even}, \\ \mathbf{Z}_2/\mathbf{Z}, & k \text{ odd}. \end{cases}$$

Furthermore,

$$(3.7) \quad \begin{aligned} \pi_0(\text{Comp}_\eta(k)^{hC_2}) &= 0, \\ \pi_1(\text{Comp}_\eta(k)^{hC_2}) &\xrightarrow{\sim} \mathbf{Z}_2^\times / \{\pm 1\} \oplus \text{RedComp}_\eta(k)^{C_2} \end{aligned}$$

by restriction.

Proof. Put $n = k - 1 \geq 1$. After choosing a point on the underlying sphere, cochains split as modules into the coefficient spectrum and its $(-n)$ -fold suspension. Completion is consequently an injection in each homotopy degree, with cokernel a finite sum of copies of $\mathbf{Z}_2/\mathbf{Z}$.

In degree zero, evaluation and scalar inclusion give

$$\begin{aligned} \pi_0 C^*(S^n; \text{KU}) &= \mathbf{Z} \oplus \pi_n \text{KU}, \\ \pi_0 C^*(S^n; \text{KU}_2^\wedge) &= \mathbf{Z}_2 \oplus \pi_n \text{KU}_2^\wedge. \end{aligned}$$

The second summands are square-zero ideals. Their product factors through the reduced stable diagonal $S^n \rightarrow S^{2n}$, whose class belongs to $\pi_{-n}\mathbb{S} = 0$. Although the chosen point need not be fixed, all evaluations agree on π_0 because the sphere is connected. Since conjugation acts trivially on the degree-zero coefficient rings, these are equivariant splittings of rings. The units of a square-zero extension split as the scalar units times the additive ideal, by $(a, x) \mapsto a(1 + x)$. The completion map is therefore injective on units, with quotient (3.6); its reduced quotient is zero for odd n , and is $\mathbf{Z}_2/\mathbf{Z}$ for even n .

Proposition 2.7 gives an isomorphism on π_0 of the two Picard spectra. Their cofiber is 1-connective, and (2.1) now gives all the asserted coefficient groups. By Lemmas 3.2 and

3.3, $(\tau_{\geq 2} \text{Comp}_{\eta}(k))^{hC_2}$ is 2-connective. The Postnikov sequence consequently reduces its fixed-point groups in degrees zero and one to

$$\begin{aligned} H^1(C_2; \mathbf{Z}_2^{\times}/\{\pm 1\} \oplus \text{RedComp}_{\eta}(k)), \\ H^0(C_2; \mathbf{Z}_2^{\times}/\{\pm 1\} \oplus \text{RedComp}_{\eta}(k)). \end{aligned}$$

The first group vanishes: its scalar part is the two-torsion subgroup, and its reduced-completion part vanishes by averaging. This proves (3.7), including the description by restriction. $\square$

Proof of completion invariance in Theorem 1.1. For $k = 1$, Lemma 2.1 proves that completion is an isomorphism. For $k \geq 2$, fixed points of (3.4) and Lemma 2.4 give

$$\begin{aligned} (3.8) \quad \pi_1(\text{pic}(\text{KU}_2^{\wedge} \otimes_{\text{KO}_2^{\wedge}} \hat{A}_k)^{hC_2}) &\longrightarrow \pi_1(\text{Comp}_{\eta}(k)^{hC_2}) \\ &\longrightarrow \text{Pic}(A_k) \longrightarrow \text{Pic}(\hat{A}_k) \longrightarrow \pi_0(\text{Comp}_{\eta}(k)^{hC_2}). \end{aligned}$$

The last group vanishes by Lemma 3.5. To prove injectivity as well, we lift both summands of the degree-one comparison group in (3.7).

A class in $\mathbf{Z}_2^{\times}/\{\pm 1\}$ has a representative $a \in \mathbf{Z}_2^{\times} = (\pi_0 \text{KO}_2^{\wedge})^{\times}$. Its image in $\pi_0 \hat{A}_k$ is a unit and hence, by descent, a class in the first group of (3.8). Restriction identifies its comparison image with the prescribed scalar class and zero reduced-completion component.

For $v \in \text{RedComp}_{\eta}(k)^{C_2}$, its unique half $w = v/2$ is also invariant. The defining cofiber sequence is onto on π_1 , since its map on π_0 is an isomorphism. Choose $y \in \pi_1 \text{pic}(\text{KU}_2^{\wedge} \otimes_{\text{KO}_2^{\wedge}} \hat{A}_k)$ mapping to $(0, w)$, and apply the transfer of Lemma 3.4. Naturality and the transfer formula identify the restriction of its comparison image with $(0, w + gw) = (0, v)$. Restriction in (3.7) is an isomorphism, so this lifts precisely the desired second summand.

The first arrow in (3.8) is therefore onto. Exactness proves the claimed isomorphism. It is the original extension-of-scalars map, so it preserves suspension and commutes with every map of the towers. $\square$

3.3. The logarithm and the relative fixed-point group.

Proposition 3.6. *For $k \geq 4$, a choice of Real Bott equivalences gives*

$$(3.9) \quad \pi_0(\widehat{\text{Rel}}_{\eta}(k)^{hC_2}) \cong \ker(\pi_{-k-2} \text{KO}_2^{\wedge} \xrightarrow{c} \pi_{-k-2} \text{KU}_2^{\wedge}) \cong \begin{cases} \mathbf{Z}/2, & k \equiv 4, 5 \pmod{8}, \\ 0, & k \not\equiv 4, 5 \pmod{8}. \end{cases}$$

Lemma 3.7 (The completed logarithm). *The Rezk logarithm gives a natural map of spectra with conjugation action*

$$\ell : \text{gl}_1(\text{KU}_2^{\wedge}) \longrightarrow \text{KU}_2^{\wedge}.$$

On π_{2m} it acts, under the usual positive-degree identification, by multiplication by $1 - 2^{m-1}$ for $m \geq 1$. Hence there are C_2 -equivariant equivalences

$$(3.10) \quad \tau_{\geq 3} \text{gl}_1(\text{KU}_2^{\wedge}) \simeq \tau_{\geq 3} \text{KU}_2^{\wedge}, \quad \tau_{\geq 4} \text{pic}(\text{KU}_2^{\wedge}) \simeq \tau_{\geq 4} \Sigma \text{KU}_2^{\wedge}.$$

Proof. The logarithm is a natural transformation $\text{gl}_1(-) \rightarrow L_{K(1)}(-)$ [Rez06, Section 1.7 and Definition 3.6]. The spectrum $\text{KU}_2^{\wedge}$ is $K(1)$ -local at two and satisfies the hypothesis of [Rez06, Section 1.8]; that section explicitly includes completed periodic complex and real K -theory. Naturality of the spectrum construction makes the resulting map equivariant for the coherent conjugation action.

Let ψ^2 denote the second Adams operation. For a reduced Bott class $\epsilon \in \widehat{\text{KU}}_2^{\wedge 0}(S^{2m})$, the square-zero formula following [Rez06, Theorem 1.9] gives

$$\ell(1 + \epsilon) = \epsilon - \frac{1}{2} \psi^2(\epsilon).$$

The group in question is free over $\mathbf{Z}_2$, so there is no torsion ambiguity in this formula. Let L be a Hopf line bundle on S^2 . Its reduced class satisfies $\psi^2(L - 1) = L^2 - 1 = 2(L - 1)$;

external products give $\psi^2(\epsilon) = 2^m \epsilon$ on S^{2m} . The positive-degree identification for units sends $1 + \epsilon$ to ϵ [Rez06, Section 1.5]. This proves the coefficient formula. For $m \geq 2$ its multiplier is odd, while the positive odd homotopy groups vanish, proving the first equivalence. Finally $\tau_{\geq 1} \text{pic}(\text{KU}_2^\wedge) \simeq \Sigma \text{gl}_1(\text{KU}_2^\wedge)$; suspending gives the second. $\square$

Lemma 3.8 (Real Bott periodicity). *For $R \in \{\text{KU}, \text{KU}_2^\wedge\}$, there is an equivalence $\Sigma^{1+\sigma} R \simeq R$ of spectra with conjugation action. Consequently, for $k \geq 0$,*

$$(3.11) \quad \Sigma^{2-k\sigma} R \simeq \Sigma^{k+2} R.$$

Proof. Restrict the genuine Real Bott equivalence to spectra with action; the Bott map and its conjugation compatibility are given in [HS14, Section 5, pp. 11–12 of the cited version]. Completion gives the same equivalence for $\text{KU}_2^\wedge$. It follows that $\Sigma^{-\sigma} R \simeq \Sigma R$; iteration and suspension by two give (3.11). $\square$

Proposition 3.9. *For $k \geq 4$, there is an equivalence in $\text{Fun}(BC_2, \text{Sp})$*

$$(3.12) \quad \widehat{\text{Rel}}_\eta(k) \simeq \tau_{\geq 1}(\Sigma^{2-k\sigma} \text{KU}_2^\wedge) \simeq \tau_{\geq 1}(\Sigma^{k+2} \text{KU}_2^\wedge).$$

Proof. The underlying dimension of $1 - k\sigma$ is $1 - k$, so

$$\tau_{\geq 1}(\Sigma^{1-k\sigma} \text{pic}(\text{KU}_2^\wedge)) \simeq \Sigma^{1-k\sigma} \tau_{\geq k} \text{pic}(\text{KU}_2^\wedge).$$

For $k \geq 4$, Lemma 3.7 identifies the right-hand side with $\Sigma^{1-k\sigma} \tau_{\geq k} \Sigma \text{KU}_2^\wedge$. Apply Proposition 2.8 and then Lemma 3.8. $\square$

Lemma 3.10. *For a spectrum E with an action of a group H , restriction induces*

$$\pi_0((\tau_{\geq 1} E)^{hH}) \cong \ker(\pi_0(E^{hH}) \rightarrow (\pi_0 E)^H).$$

Proof. Take fixed points of $\tau_{\geq 1} E \rightarrow E \rightarrow \tau_{\leq 0} E$. The space $\Omega^\infty \tau_{\leq 0} E$ is discrete, with set of components $\pi_0 E$. Since Ω^∞ preserves limits, the fixed-point spectrum of $\tau_{\leq 0} E$ has zero π_1 and $\pi_0 = (\pi_0 E)^H$. The long exact sequence gives the assertion, without a boundedness or spectral-sequence convergence hypothesis. $\square$

Proof of Proposition 3.6. Apply Lemma 3.10 to $E = \Sigma^{k+2} \text{KU}_2^\wedge$, using Proposition 3.9. Galois descent gives $(\text{KU}_2^\wedge)^{hC_2} \simeq \text{KO}_2^\wedge$, and integer suspensions commute with limits of spectra. The restriction map is complexification into $(\pi_{-k-2} \text{KU}_2^\wedge)^{C_2}$. Including invariants into the full coefficient group leaves its kernel unchanged.

The completed Wood sequence identifies the kernel of complexification in degree d with the image of $\eta : \pi_{d-1} \text{KO}_2^\wedge \rightarrow \pi_d \text{KO}_2^\wedge$. By (2.2), this image is $\mathbf{Z}/2$ exactly for $d \equiv 1, 2 \pmod{8}$, and is zero otherwise. Substituting $d = -k - 2$ gives the formula. $\square$

Remark 3.11. With the usual generators of $\pi_* \text{KO}_2^\wedge$, the nonzero classes in (3.9) correspond to

$$\eta^2 \beta^{-(m+1)} \quad (k = 8m + 4), \quad \eta \beta^{-(m+1)} \quad (k = 8m + 5), \quad m \geq 0.$$

Their degrees are respectively $-8m - 6$ and $-8m - 7$. The logarithm equivalence used here is a completed equivalence; its multiplier need not be an integral unit. Moreover the Bott choices compute individual groups and do not identify the original tower maps with unspecified maps between integer suspensions. We retain the cofiber sequences (2.8) for naturality.

Lemma 3.12. *Let $f : E \rightarrow E'$ be a map of 1-connective spectra with C_2 -action. If every $\pi_i f$ is injective with uniquely two-divisible cokernel, then f induces an isomorphism*

$$\pi_0(E^{hC_2}) \xrightarrow{\sim} \pi_0((E')^{hC_2}).$$

Proof. Its cofiber T is 1-connective and has uniquely two-divisible homotopy groups. Lemma 3.3 gives $\pi_0(T^{hC_2}) = 0$ and identifies $\pi_1(T^{hC_2})$ with $(\pi_1 T)^{C_2}$. For an invariant $v \in \pi_1 T$, its unique half w is invariant. Since $\pi_0 E = 0$, choose a lift $y \in \pi_1 E'$ of w . Its transfer maps to v by Lemma 3.4 and the restriction identification. Thus $\pi_1((E')^{hC_2}) \rightarrow \pi_1(T^{hC_2})$ is onto, and the long exact sequence gives the claimed isomorphism. $\square$

Proposition 3.13. *For every $k \geq 2$, completion induces a natural isomorphism*

$$\pi_0(\mathrm{Rel}_\eta(k)^{hC_2}) \xrightarrow{\sim} \pi_0(\widehat{\mathrm{Rel}}_\eta(k)^{hC_2}).$$

For $k \geq 4$, these groups are given by (3.9).

Proof. For $i \geq 1$ and $k \geq 2$, Proposition 2.8 identifies the underlying coefficient map with

$$\pi_i \mathrm{Rel}_\eta(k) = \pi_{i+k-2} \mathrm{KU} \longrightarrow \pi_{i+k-2} \mathrm{KU}_2^\wedge = \pi_i \widehat{\mathrm{Rel}}_\eta(k).$$

The Picard index $i + k - 1$ is at least two, so (2.1) applies even in the lowest degree. The displayed map is either $\mathbf{Z} \hookrightarrow \mathbf{Z}_2$ or the map between zero groups. Lemmas 3.2 and 3.12 prove the isomorphism. It is induced by the original completion square and hence respects the relative towers. Proposition 3.6 gives the formula for $k \geq 4$. $\square$

3.4. The boundary and the suspension summand. For $k \geq 4$, taking fixed points of (2.8), which is exact in spectra, gives

$$(3.13) \quad \mathrm{Pic}(\mathrm{KO}) \xrightarrow{i_k} \mathrm{Pic}(A_k) \xrightarrow{q_k} \pi_0(\mathrm{Rel}_\eta(k)^{hC_2}) \xrightarrow{\delta_k} \pi_{-1}(\mathrm{pic}(\mathrm{KU})^{hC_2}).$$

The rightmost group is a negative homotopy group of the untruncated homotopy fixed points, not of the connective spectrum $\mathrm{pic}(\mathrm{KU})$.

Proposition 3.14. *For $k \geq 4$, extension of scalars embeds $\mathrm{Pic}(\mathrm{KO})$ as a split suspension subgroup of $\mathrm{Pic}(A_k)$. For $k \geq 6$, there is an isomorphism*

$$(3.14) \quad \begin{aligned} (r_k, q_k) : \mathrm{Pic}(A_k) &\xrightarrow{\sim} \mathrm{Pic}(\mathrm{KO}) \oplus \pi_0(\mathrm{Rel}_\eta(k)^{hC_2}), \\ r_k &= i_4^{-1} \circ (\mathrm{Pic}(A_k) \rightarrow \mathrm{Pic}(A_4)). \end{aligned}$$

For $k = 4, 5$, the boundary δ_k is injective on $\pi_0(\mathrm{Rel}_\eta(k)^{hC_2}) = \mathbf{Z}/2$.

Proof. Lemma 2.1 and Proposition 3.1 show that i_4 is an isomorphism: both groups are cyclic of order eight and extension of scalars preserves their suspension generators. The structural maps of the tower give $r_k i_k = \mathrm{id}$, proving the split injection.

Proposition 3.13 gives $\pi_0(\mathrm{Rel}_\eta(6)^{hC_2}) = 0$. For $k \geq 6$, the map $A_k \rightarrow A_6$ induces a map of the relative cofiber sequences with identity on $\mathrm{pic}(\mathrm{KU})$. Naturality of the connecting homomorphisms makes δ_k factor through this vanishing sixth-stage group, so $\delta_k = 0$. Thus (3.13) is short exact for $k \geq 6$. The retraction r_k splits it explicitly: given a lift x of a relative class, $x - i_k(r_k(x))$ lies in $\ker r_k$ and has the same relative image. Since $\ker r_k \cap \mathrm{im} i_k = 0$, the restriction $q_k|_{\ker r_k}$ is an isomorphism. This proves (3.14). At $k = 4, 5$, the map i_k is already an isomorphism by Proposition 3.1, so exactness forces δ_k to be injective. $\square$

Proof of the group formula in Theorem 1.1. The first five cases are Proposition 3.1. For $k \geq 6$, combine Proposition 3.14 with the relative-group formula of Proposition 3.13 and $\mathrm{Pic}(\mathrm{KO}) = \mathbf{Z}/8$. The first integers in this range congruent to four or five modulo eight are twelve and thirteen. The suspension summand is the image of i_k ; its complement is $\ker r_k$, which is precisely the kernel of the map to $\mathrm{Pic}(A_4)$. Theorem 1.1 gives the completed formula and the same canonical decomposition, with naturality along the tower. $\square$

Corollary 3.15. *For every $m \geq 1$, the maps*

$$\mathrm{Pic}(A_{8m+6}) \longrightarrow \mathrm{Pic}(A_{8m+5}), \quad \mathrm{Pic}(\widehat{A}_{8m+6}) \longrightarrow \mathrm{Pic}(\widehat{A}_{8m+5})$$

have image the suspension subgroup of order eight and cokernel $\mathbf{Z}/2$. In particular the maps from stage fourteen to stage thirteen are not surjective.

Proof. The source is cyclic of order eight on suspension; the target is $\mathbf{Z}/8 \oplus \mathbf{Z}/2$, with the same suspension subgroup. Extension of scalars sends suspension to suspension, which determines the image. Naturality of completion gives the completed assertion. $\square$

4. LOCAL MODULES AND THE COMPLETED ETA-POWER TOWER

Completion of a coefficient ring and completion of its module category have different effects on invertibility. We first identify the local systems realized by complete cochain modules. This also supplies the realizability criterion for finite coefficients used in Section 6.

Lemma 4.1. *On KO-modules, $K(1)$ -localization at two is derived two-completion. Every module over $KO/2^m$ or $KO/(2^m, \eta^n)$ is already $K(1)$ -local. Its ordinary relative tensor products are therefore local as well.*

Proof. For a KU-module N , the spectrum $N/2$ is a module over $KU/2$, whose Bousfield class is that of $K(1)$ at two. Such a module is local: if Z is $K(1)$ -acyclic, the free-forgetful adjunction identifies maps from Z to that module with maps from $(KU/2) \otimes Z$, which is zero. Each $N/2^j$ is a successive extension of such modules, so $\mathrm{Rlim}_j N/2^j$ is local. The completion map is a mod-two equivalence; its cofiber is uniquely two-divisible and hence $K(1)$ -acyclic. It is consequently the localization map.

Complexification of a KO-module is its cofiber by η , by the Wood sequence. It commutes with completion and preserves locality. Conversely, faithful C_2 -Galois descent recovers a KO-module from its complexification by homotopy fixed points. Complete spectra and local spectra are closed under limits. These observations prove the first assertion; the Galois and Wood inputs are [Rog08, Proposition 5.3.1 and (5.3.2)].

Multiplication by 2^m is null on a module over either displayed quotient. The inverse system computing maps from $\mathbb{S}[1/2]$ into this module has a uniformly null iterate. Both its limit and its derived limit vanish, so the module is derived two-complete. The first assertion applies also to relative tensor products, which remain modules over the quotient. $\square$

For a monoidal category of modules, $\mathcal{P}$ denotes its Picard space. We use the coefficient calculations

$$(4.1) \quad \mathrm{Pic}_{K(1)}(KU_2^\wedge) = \mathrm{Pic}(KU_2^\wedge) = \mathbf{Z}/2, \quad \mathrm{Pic}_{K(1)}(KO_2^\wedge) = \mathbf{Z}/8, \quad \mathrm{Pic}(KU/2^m) = \mathbf{Z}/2.$$

The local calculations are [LZ25, Main Theorem E]; the ordinary completed calculation follows from [MS16, Theorem 2.4.6]. For the specified Moore quotient structures, the finite calculation is [LLZ26, Theorem 2.25 and Example 2.27]. In every case the generator is suspension. The ordinary and local Picard spaces of $KU_2^\wedge$ are equivalent: the comparison is an isomorphism on components by (4.1), and it identifies automorphism spaces because the local module category is full.

Lemma 4.2 (Cochain modules and completion). *Let T be $KU_2^\wedge$ or one of the specified $\mathbb{E}_2$ -algebras $KU/2^m$, and let X be a finite CW space. Evaluation induces a fully faithful exact monoidal functor*

$$\mathcal{L} : \mathrm{Mod}_{C^*(X;T)} \longrightarrow \mathrm{Fun}(X, \mathrm{Mod}_T)$$

whose essential image is the localizing subcategory generated by the constant unit. Its right adjoint is the section functor Γ_X .

For $T = KU_2^\wedge$, completing both module categories gives a fully faithful functor

$$\widehat{\mathcal{L}} : (\mathrm{Mod}_{C^*(X;KU_2^\wedge)})_2^\wedge \longrightarrow \mathrm{Fun}(X, (\mathrm{Mod}_{KU_2^\wedge})_2^\wedge).$$

A complete local system belongs to its image if and only if its reduction modulo two belongs to the ordinary localizing subcategory generated by the unit. These comparisons are natural in X and in the coefficients, and respect semilinear finite-group actions.

Proof. The endomorphism ring of the constant unit is $C^*(X;T)$. A finite CW presentation of X computes sections by a finite homotopy limit of evaluation functors. Finite limits commute with colimits in stable categories, so Γ_X preserves colimits and the constant unit is compact. The unit of the extension-section adjunction is an equivalence on $C^*(X;T)$ and hence, by colimits and cofibers, on all $C^*(X;T)$ -modules. This proves full faithfulness and

identifies the essential image. Restriction to compact objects recovers the perfect-cochain comparison of [MS16, Section 4.2]. Evaluation is monoidal also for the merely monoidal module categories under consideration.

The section functor commutes with limits and finite cofibers. For a complete M we therefore have

$$\Gamma_X \widehat{\mathcal{L}}(M) \simeq \operatorname{Rlim}_j \Gamma_X \mathcal{L}(M/2^j) \simeq \operatorname{Rlim}_j M/2^j \simeq M.$$

This proves completed full faithfulness. Reducing the counit on a complete local system modulo two gives the ordinary Morita counit on its reduction. It is an equivalence precisely under the condition in the statement. A map between complete modules is an equivalence when its reduction modulo two is one. The constructions are functorial, giving the asserted compatibilities. $\square$

Proposition 4.3 (Spherical realizability). *Let T be $\mathrm{KU}_2^\wedge$ or a specified $\mathrm{KU}/2^m$ with $m \geq 5$, as in Lemma 4.2. After suspension, an invertible local system on S^d , $d \geq 1$, has fiber T . On the circle it is classified by a unit $a \in (\pi_0 T)^\times$. On an even-dimensional sphere it is constant. On an odd-dimensional sphere of dimension at least three it is classified by*

$$au^{(d-1)/2} \in \pi_{d-1} T, \quad |u| = 2.$$

For complete $\mathrm{KU}_2^\wedge$ -cochain modules the realizable classes are all $a \in \mathbf{Z}_2^\times$ on the circle and exactly $a \in 2\mathbf{Z}_2$ on a higher odd sphere. For ordinary $(\mathrm{KU}/2^m)$ -cochain modules they are all $a \in (\mathbf{Z}/2^m)^\times$ on the circle and exactly $a \in 2(\mathbf{Z}/2^m)$ on a higher odd sphere. In both cases the Picard-space comparison is an inclusion of components and an equivalence on positive homotopy groups.

Proof. The classification follows from (4.1) and

$$\pi_1 \mathcal{P}(T) = (\pi_0 T)^\times, \quad \pi_j \mathcal{P}(T) = \pi_{j-1} T \quad (j \geq 2).$$

Local systems on a connected based space are modules over $T[\Omega X]$. For odd $d \geq 3$, the James equivalence [Jam55, Theorems 1.11 and 5.6] identifies $T[\Omega S^d]$ with the free associative T -algebra on a class y of degree $d-1$. Set $x = u^{-(d-1)/2}y$. This degree-zero generator gives the central polynomial algebra $T[x]$, and the constant unit is $T[x]/x$.

The localizing subcategory generated by $T[x]/x$ is exactly the subcategory of x -torsion modules. Localization at x kills the generator. Conversely, the fiber of $M \rightarrow M[x^{-1}]$ is a filtered colimit of shifts of M/x^j , and the finite x -filtration builds each of these from $T \otimes_{T[x]} M$. For the line local system above, y acts by $au^{(d-1)/2}$, so x acts by a . Its localization vanishes exactly when a is nilpotent. In $\mathbf{Z}/2^m$ this means that a is even. In the completed category, Lemma 4.2 imposes nilpotence only after reduction modulo two, again giving even a .

For $d = 1$, the augmentation element $z - 1$ in $T[z, z^{-1}]$ acts by $a - 1$, which is nilpotent modulo 2^m , respectively zero modulo two, for every unit in the stated coefficient rings. The inverse local system satisfies the same criterion. Monoidal full faithfulness therefore makes each realized preimage invertible. In the ordinary category it is automatically perfect, since an invertible module is compact. Full faithfulness identifies automorphism spaces, proving the last assertion. $\square$

In the ordinary $\mathrm{KU}_2^\wedge$ -module category, the same argument requires actual nilpotence in $\mathbf{Z}_2$. It allows only monodromy 1 on the circle and clutching coefficient 0 on a higher odd sphere. This is the distinction between the ordinary calculation and Theorem 1.2.

The geometric comparison of Proposition 2.3, after completion, identifies complexification of $\widehat{A}_k$ with $C^*(S(k\sigma); \mathrm{KU}_2^\wedge)$. Lemma 4.1 shows that Galois descent restricts to local module categories. Since the antipodal sphere is free, the Picard group of $\widehat{A}_k$ in that category embeds into

$$(4.2) \quad \pi_0 \Gamma(\mathbf{RP}^{k-1}; \operatorname{pic}(\mathrm{KU}_2^\wedge)),$$

where the coefficient local system has monodromy conjugation. Its image consists exactly of sections whose pullback to S^{k-1} satisfies Proposition 4.3. This criterion includes coherent descent: an equivariant object in a stable full essential image lifts its action through the fully faithful comparison.

4.1. The first five stages.

Lemma 4.4. *For $k = 1, 2, 3, 4, 5$, respectively, the local Picard groups are*

$$\mathbf{Z}/2, \quad \mathbf{Z}/4 \oplus \mathbf{Z}_2, \quad \mathbf{Z}/4, \quad \mathbf{Z}/8, \quad \mathbf{Z}/8.$$

The finite factors are generated by suspension.

Proof. The first case is (4.1). Over $\mathbf{RP}^1$, the kernel of parity is identified with $\mathbf{Z}_2^\times$ by monodromy. Under this identification, twice the suspension class corresponds to the unit -1 : conjugation negates the Bott trivialization of $\Sigma^2 \mathrm{KU}_2^\wedge$. All circular local systems are admissible, by Proposition 4.3. Since

$$\mathbf{Z}_2^\times = \{\pm 1\} \times (1 + 4\mathbf{Z}_2), \quad 1 + 4\mathbf{Z}_2 \cong \mathbf{Z}_2,$$

the second group is $\mathbf{Z}/4 \oplus \mathbf{Z}_2$.

The finite skeletal section sequence is

$$E_2^{s,t} = H^s(\mathbf{RP}^{k-1}; \pi_t \mathrm{pic}(\mathrm{KU}_2^\wedge)) \implies \pi_{t-s} \Gamma(\mathbf{RP}^{k-1}; \mathrm{pic}(\mathrm{KU}_2^\wedge)), \quad 0 \leq s \leq k-1.$$

Only the Postnikov section through degree $k-1$ affects its zero stem, so its filtration is finite. For $\mathbf{RP}^2$ the zero diagonal has only $\mathbf{Z}/2$ in degree zero and $H^1(\mathbf{RP}^2; \mathbf{Z}_2^\times) = \{\pm 1\}$. Thus its order is at most four, and restriction to stage two detects a suspension of order four.

For $\mathbf{RP}^3$ and $\mathbf{RP}^4$, the only additional diagonal term is

$$H^3(\mathbf{RP}^N; \mathbf{Z}_2^{\mathrm{sign}}) = \mathbf{Z}/2 \quad (N = 3, 4),$$

as follows from the cellular differentials $2, 0, 2, 0, \dots$ for sign coefficients. The order is therefore at most eight. The map $\mathrm{KO}_2^\wedge \rightarrow \hat{A}_4$ induces an isomorphism on π_4 : its fiber is $\Sigma^4 \mathrm{KO}_2^\wedge$, its attaching map is $\eta^4 = 0$, and $\pi_{-1} \mathrm{KO}_2^\wedge = 0$. Complexification sends a generator in degree four to $2u^2$, which is not a unit. Hence the suspension has order greater than four. The real Bott unit makes its order eight, and restriction from stage five to stage four proves the final case. $\square$

4.2. The relative lattice and its boundary. Fix $k \geq 5$. Let $\mathrm{pic}^{\mathrm{loc}}(C^*(S(k\sigma); \mathrm{KU}_2^\wedge))$ denote the Picard spectrum of the symmetric monoidal category of local modules. In $\mathrm{Fun}(BC_2, \mathrm{Sp})$ put

$$\mathrm{Rel}_k^{\mathrm{loc}} = \mathrm{cofib}(\mathrm{pic}(\mathrm{KU}_2^\wedge) \longrightarrow \mathrm{pic}^{\mathrm{loc}}(C^*(S(k\sigma); \mathrm{KU}_2^\wedge))).$$

The representation-sphere sequence and evaluation induce an equivariant map

$$(4.3) \quad \mathrm{Rel}_k^{\mathrm{loc}} \longrightarrow \Sigma^{1-k\sigma} \mathrm{pic}(\mathrm{KU}_2^\wedge).$$

It is an isomorphism on positive homotopy groups. Its degree-zero image is the admissible clutching subgroup in Proposition 4.3. These assertions follow from the long exact sequences, using evaluation at a point only after forgetting the action. The map itself requires no equivariant basepoint.

For a spectrum E with C_2 -action, define

$$\mathrm{Lat}_2(E) = \mathrm{fib}(\tau_{\geq 0} E \rightarrow H(\pi_0 E/2)).$$

Lemma 4.5. *For $k \geq 5$ there is an equivalence of spectra with C_2 -action*

$$(4.4) \quad \mathrm{Rel}_k^{\mathrm{loc}} \simeq \mathrm{Lat}_2(\Sigma^{k+2} \mathrm{KU}_2^\wedge).$$

For any spectrum E with C_2 -action,

$$(4.5) \quad \pi_0(\mathrm{Lat}_2(E)^{hC_2}) \cong \ker(\pi_0(E^{hC_2}) \rightarrow (\pi_0 E/2)^{C_2}).$$

Proof. The equivariant logarithm comparison of Lemma 3.7 gives

$$\tau_{\geq 4} \text{pic}(\text{KU}_2^\wedge) \simeq \tau_{\geq 4} \Sigma \text{KU}_2^\wedge.$$

Its coefficient multipliers in the relevant degrees are two-adic units. The Real Bott equivalence of Lemma 3.8 gives $\Sigma^{2-k\sigma} \text{KU}_2^\wedge \simeq \Sigma^{k+2} \text{KU}_2^\wedge$.

The nonnegative part of $\Sigma^{1-k\sigma} \text{pic}(\text{KU}_2^\wedge)$ depends only on $\tau_{\geq k-1} \text{pic}(\text{KU}_2^\wedge)$. Thus $k \geq 5$ places all relevant degrees in the logarithm range. In degree zero the admissible subgroup is twice the coefficient group when k is even; the coefficient group is zero when k is odd. The logarithm and Bott identifications preserve these subgroups, since their coefficient multipliers are units. Applying this to (4.3) proves (4.4).

The cofiber of $\text{Lat}_2(E) \rightarrow E$ is coconnective, with degree-zero group $\pi_0 E/2$. Its infinite loop space is discrete. Its homotopy fixed points consequently have no degree-one homotopy and have degree-zero group $(\pi_0 E/2)^{C_2}$. The long exact sequence proves (4.5). $\square$

Corollary 4.6. *For $k \geq 5$,*

$$\pi_0((\text{Rel}_k^{\text{loc}})^{hC_2}) \cong \ker(\pi_{-k-2} \text{KO}_2^\wedge \xrightarrow{c \bmod 2} (\pi_{-k-2} \text{KU}_2^\wedge)/2).$$

In residues 0, 1, 2, 3, 4, 5, 6, 7 of k modulo eight, these groups are respectively

$$0, \quad 0, \quad \mathbf{Z}_2, \quad 0, \quad \mathbf{Z}/2, \quad \mathbf{Z}/2, \quad 2\mathbf{Z}_2, \quad 0.$$

Proof. Apply Lemma 4.5 and $(\text{KU}_2^\wedge)^{hC_2} \simeq \text{KO}_2^\wedge$. The coefficient ring and complexification are

$$\begin{aligned} \pi_* \text{KO}_2^\wedge &= \mathbf{Z}_2[\eta, \alpha, \beta^{\pm 1}]/(2\eta, \eta^3, \eta\alpha, \alpha^2 - 4\beta), & |\eta| &= 1, \quad |\alpha| = 4, \quad |\beta| = 8, \\ c(\alpha) &= 2u^2, & c(\beta) &= u^4, \end{aligned}$$

and complexification kills torsion [Rog08, Section 5.3]. Reduction modulo two gives the table. $\square$

Taking homotopy fixed points of the defining relative cofiber sequence gives

$$(4.6) \quad \text{Pic}_{K(1)}(\text{KO}_2^\wedge) \longrightarrow \text{Pic}_{K(1)}(\hat{A}_k) \longrightarrow \pi_0((\text{Rel}_k^{\text{loc}})^{hC_2}) \xrightarrow{\delta_k^{\text{loc}}} \pi_{-1}(\text{pic}(\text{KU}_2^\wedge)^{hC_2}).$$

The group on the right is the negative homotopy group of the untruncated homotopy fixed-point spectrum, not of the connective Picard spectrum of $\text{KO}_2^\wedge$.

Lemma 4.7. *For $k \geq 6$, the boundary δ_k^{loc} is zero. For $k \geq 5$, restriction to stage five splits the suspension subgroup.*

Proof. Lemma 4.4 identifies the map $\text{Pic}_{K(1)}(\text{KO}_2^\wedge) \rightarrow \text{Pic}_{K(1)}(\hat{A}_5)$ with an isomorphism. Restriction to this stage is a retraction at every later stage. For $k \geq 7$, the map to stage seven and naturality of (4.6) show that δ_k^{loc} factors through $\pi_0((\text{Rel}_7^{\text{loc}})^{hC_2}) = 0$.

At stage six, consider sections over $\mathbf{RP}^5$ of $\tau_{\geq 5} \text{pic}(\text{KU}_2^\wedge)$. Their component group is

$$H^5(\mathbf{RP}^5; \pi_5 \text{pic}(\text{KU}_2^\wedge)) = H^5(\mathbf{RP}^5; \mathbf{Z}_2) = \mathbf{Z}_2.$$

No higher obstruction occurs on this five-dimensional space. Both S^5 and $\mathbf{RP}^5$ are orientable and the covering has degree two, so pullback sends these classes onto $2\mathbf{Z}_2$. They are all admissible by Proposition 4.3. By Corollary 4.6, the restriction of $\pi_0((\text{Rel}_6^{\text{loc}})^{hC_2})$ to $\pi_0 \text{Rel}_6^{\text{loc}}$ identifies it with the same subgroup. The logarithm is a unit on $\pi_5 \text{pic}(\text{KU}_2^\wedge)$. Hence these top-cell sections lift every element of $\pi_0((\text{Rel}_6^{\text{loc}})^{hC_2})$, proving $\delta_6^{\text{loc}} = 0$. $\square$

Proof of Theorem 1.2. Lemma 4.4 gives the first five stages. For $k \geq 6$, (4.6) and Lemma 4.7 give a split exact sequence with kernel $\mathbf{Z}/8$ and quotient $\pi_0((\text{Rel}_k^{\text{loc}})^{hC_2})$. Corollary 4.6 gives the stated formula. The complementary subgroup is canonically the kernel of restriction to stage five. At stage five the relative group $\pi_0((\text{Rel}_5^{\text{loc}})^{hC_2}) = \mathbf{Z}/2$ contributes no absolute class: the small-stage calculation forces its boundary to be injective. $\square$

5. THE COEFFICIENT ALGEBRA $KO/32$

Fix the $\mathbb{E}_2$ -structure on $\mathbb{S}/32$ obtained by realizing Burklund's $\mathbf{F}_2$ -synthetic quotient construction [Bur22, Theorem 3.2]. All quotient algebras in this section carry its base changes. Completion does not change the finite quotients, so we use $KO_2^\wedge/32$ and $KU_2^\wedge/32$ in the proof, with the Bott generator and conjugation conventions of (2.2).

Theorem 5.1. *For this quotient structure, suspension induces an isomorphism*

$$\mathbf{Z}/8 \xrightarrow{\sim} \text{Pic}(KO_2^\wedge/32).$$

In its Picard descent sequence, the character $\chi_{17}: C_2 \rightarrow (\mathbf{Z}/32)^\times$, defined by $\chi_{17}(g) = 17$, satisfies

$$(5.1) \quad d_3(\chi_{17}) = [16u] \neq 0 \quad \text{in } H^4(C_2; (\mathbf{Z}/32)u).$$

The descent calculation of Levy–Li–Zhang leaves one possible line beyond the suspension subgroup, with character 17. We first refine the ordinary Bockstein to a coherent associative derivation. We then show that the underlying $KO_2^\wedge$ -module of such a line permits only even multiples of the complex Bockstein to descend. An invertible module admits a connection for our derivation, and that connection complexifies to the Bockstein itself. This excludes the line. The separate reduction to exponent 16 at the end of the section identifies the differential; this identification is needed on the finite projective skeletons used in Section 6.

5.1. The remaining character. The faithful C_2 -Galois extension $KO_2^\wedge \rightarrow KU_2^\wedge$ [Rog08, Proposition 5.3.1], after base change by the chosen Moore algebra, gives the descent sequence

$$(5.2) \quad E_2^{s,t} = H^s(C_2; \pi_t \mathcal{P}(KU_2^\wedge/32)) \implies \pi_{t-s} \mathcal{P}(KO_2^\wedge/32), \quad t - s \geq 0.$$

Our convention is $d_r: (s, t) \mapsto (s + r, t + r - 1)$. A differential leaving the zero stem is interpreted as a Postnikov obstruction to descent. The convergence and the finite zero-stem filtration are those in [LLZ26, Theorem 3.1, proof of Theorem 3.23, and Remark 3.31].

Lemma 5.2. *In (5.2), the characters χ_{-1} , defined by $\chi_{-1}(g) = -1$, and χ_{17} generate $H^1(C_2; (\mathbf{Z}/32)^\times) \cong (\mathbf{Z}/2)^2$. Apart from χ_{17} , the only zero-stem classes that can survive are the suspension classes in filtrations 0, 1, 3. The last two are represented by $\Sigma^2(KO_2^\wedge/32)$ and $\Sigma^4(KO_2^\wedge/32)$. Every nonsuspension class can therefore be represented by a line whose complexification is free and whose character on π_0 is 17. The only possible nonzero outgoing differentials on χ_{17} are d_3 and d_5 , with respective targets*

$$H^4(C_2; (\mathbf{Z}/32)u) = \mathbf{Z}/2, \quad H^6(C_2; (\mathbf{Z}/32)u^2) = \mathbf{Z}/2.$$

Proof. The solutions to $a^2 = 1$ in $(\mathbf{Z}/32)^\times$ are 1, -1 , 17, -17 . The remaining assertions are precisely the part of [LLZ26, proof of Theorem 3.23, Figure 1(b), and Remark 3.31] that precedes the exclusion of the exceptional character. Here $\text{Pic}(KU_2^\wedge/32) = \mathbf{Z}/2$, $\pi_1 \mathcal{P}(KU_2^\wedge/32) = (\mathbf{Z}/32)^\times$, and $\pi_t \mathcal{P}(KU_2^\wedge/32) = \pi_{t-1}(KU_2^\wedge/32)$ for $t \geq 2$. The latter coefficient action on u^j is $(-1)^j$. The filtration-three suspension class is the reduction of that represented by $\Sigma^4 KO_2^\wedge$. Subtracting the three suspension classes gives the normalization asserted in the lemma. The degree calculation for the two possible targets is the one in [LLZ26, Proposition 3.26]; its nonvanishing conclusion there is used only for exponents at least 64. $\square$

5.2. A coherent Bockstein. For a cofiber $E/32$, let $i: E \rightarrow E/32$ and $\partial: E/32 \rightarrow \Sigma E$ be its cofiber maps. Its ordinary Bockstein is

$$Q_{E/32}: E/32 \xrightarrow{\partial} \Sigma E \xrightarrow{\Sigma i} \Sigma(E/32).$$

The following normalized obstruction calculation will also construct the reduction map needed later.

Lemma 5.3. *Let A be a unital $\mathbb{E}_1$ -algebra in a presentably symmetric monoidal stable category, and let N be an A -bimodule. Put $\overline{A} = \text{cofib}(\mathbf{1} \rightarrow A)$. Suppose $D: A \rightarrow N$ is equipped with a nullhomotopy on the unit. Once the lower coherences have been chosen, the arity- r obstruction to extending (id, D) to a unital associative section of $A \oplus N \rightarrow A$ lies in*

$$[\Sigma^{r-2}\overline{A}^{\otimes r}, N], \quad r \geq 2.$$

For a prescribed unit-preserving underlying map $A \rightarrow A'$, the analogous obstruction lies in $[\Sigma^{r-2}\overline{A}^{\otimes r}, A']$.

Proof. The unital A_∞ mapping obstruction theory is described in [Ang08, p. 1003]. The arity- r multiplihedron is a disk of dimension $r - 1$. Its boundary is fixed by the lower coherences, and the unit condition fixes the homotopy fat wedge where at least one input is the unit. The quotient of this extension pair is $\Sigma^{r-1}\overline{A}^{\otimes r}$. Its desuspension is the source of the obstruction. For a section, work over the fixed identity map of A ; the remaining extension takes values in the fiber N of the square-zero projection. Compatible fillings in every arity give the required unital A_∞ map. $\square$

Work in the $\mathbf{F}_2$ -synthetic category of Burklund, and let ν denote the synthetic analogue functor. We use his coordinates

$$S^{w,s} = \Sigma^{-s}\nu(\Sigma^{w+s}\mathbb{S}_2^\wedge), \quad \pi_{w,s}X = [S^{w,s}, X].$$

Categorical suspension has bidegree $(1, -1)$. The deformation parameter is $\tau: S^{0,-1} \rightarrow \nu\mathbb{S}_2^\wedge$, and the synthetic lift of multiplication by two is $\tilde{2}: S^{0,1} \rightarrow \nu\mathbb{S}_2^\wedge$. The unit cofiber of $\nu\mathbb{S}_2^\wedge/\tilde{2}^5$ is $S^{1,4}$, so the boundary gives a unit-normalized map

$$(5.3) \quad \nu\mathbb{S}_2^\wedge/\tilde{2}^5 \longrightarrow S^{1,4} \longrightarrow S^{1,4} \otimes (\nu\mathbb{S}_2^\wedge/\tilde{2}^5).$$

The target has the shifted regular bimodule structure.

Lemma 5.4. *One has*

$$\pi_{1,4}\nu\mathbb{S}_2^\wedge = 0, \quad \pi_{1,4}(\nu\mathbb{S}_2^\wedge/\tilde{2}^5) = 0.$$

Proof. Let $\mathcal{A}$ be the mod-two Steenrod algebra and let ξ_i be the Milnor generators of its dual. In the normalized cobar complex, a word of length s and internal degree $s + 1$ has one entry ξ_1^2 and all other entries ξ_1 . Both entries are primitive. The identities

$$d[\xi_2] = [\xi_1^2 \mid \xi_1], \quad d[\xi_1^3] = [\xi_1^2 \mid \xi_1] + [\xi_1 \mid \xi_1^2]$$

make every such word a boundary for $s \geq 2$, by replacing an adjacent pair containing its ξ_1^2 entry. Thus $\text{Ext}_{\mathcal{A}}^{s,s+1}(\mathbf{F}_2, \mathbf{F}_2) = 0$ for $s \geq 2$. In the coordinates of [BHS23], which send (w, s) to $(w, w + s)$, [BHS23, Corollary 9.13] gives

$$\pi_{1,5+j}(\text{cofib}(\tau) \otimes \nu\mathbb{S}_2^\wedge) = \text{Ext}_{\mathcal{A}}^{4+j,5+j}(\mathbf{F}_2, \mathbf{F}_2) = 0 \quad (j \geq 0).$$

The mod-two Adams sequence of the complete sphere converges strongly, and the sphere is $H\mathbf{F}_2$ -nilpotent complete. Therefore [BHS23, Corollary 9.20] also gives vanishing for the synthetic sphere, including possible τ -torsion. This is the first assertion.

The defining cofiber sequence gives

$$\pi_{1,4}\nu\mathbb{S}_2^\wedge \longrightarrow \pi_{1,4}(\nu\mathbb{S}_2^\wedge/\tilde{2}^5) \longrightarrow \pi_{0,0}\nu\mathbb{S}_2^\wedge \xrightarrow{\tilde{2}^5} \pi_{0,5}\nu\mathbb{S}_2^\wedge.$$

The last map is injective: its source is $\mathbf{Z}_2$, identified with its τ -inversion, and its realization is multiplication by 32 [Pst23, Corollary 4.38 and Proposition 4.40]. Exactness proves the second assertion. $\square$

Proposition 5.5. *The map $Q_{\text{KO}_2^\wedge/32}$ underlies a coherent associative derivation relative to $\text{KO}_2^\wedge$. Equivalently, there is an $\mathbb{E}_1$ - $\text{KO}_2^\wedge$ -algebra section*

$$\text{KO}_2^\wedge/32 \longrightarrow \text{KO}_2^\wedge/32 \oplus \Sigma(\text{KO}_2^\wedge/32)$$

whose underlying map is $(\text{id}, Q_{\text{KO}_2^\wedge/32})$.

Proof. Apply Lemma 5.3 to (5.3). The arity- r obstruction group is

$$\begin{aligned} [\Sigma^{r-2}(S^{1,4})^{\otimes r}, S^{1,4} \otimes (\nu \mathbb{S}_2^\wedge / \tilde{2}^5)] \\ = \pi_{2r-3, 3r-2}(\nu \mathbb{S}_2^\wedge / \tilde{2}^5). \end{aligned}$$

Burklund's vanishing line is $\pi_{w,s} = 0$ for $s > \frac{1}{2}w + 5$ [Bur22, Lemma 3.3]. It applies for $r \geq 3$, since $3r - 2 > r + 7/2$. The binary obstruction is zero by Lemma 5.4. We may therefore choose all coherences. The exact symmetric monoidal functor inverting τ [Pst23, Proposition 4.40] carries the section to the ordinary Bockstein section of $\mathbb{S}/32$. Base change gives the claimed derivation; its complexification is $Q_{\mathrm{KU}_2^\wedge/32}$. $\square$

5.3. The underlying module of the exceptional line. Define a $\mathrm{KO}_2^\wedge$ -module, without a quotient-module structure, by

$$(5.4) \quad \mathrm{KU}_2^\wedge \xrightarrow{9+7g} \mathrm{KU}_2^\wedge \xrightarrow{q} \mathrm{Cof}_{17} \xrightarrow{p} \Sigma \mathrm{KU}_2^\wedge.$$

Lemma 5.6. *There is a $\mathrm{KU}_2^\wedge$ -linear equivalence $\mathrm{KU}_2^\wedge \otimes_{\mathrm{KO}_2^\wedge} \mathrm{Cof}_{17} \simeq \mathrm{KU}_2^\wedge/32$ with descent character 17 on π_0 . Moreover,*

$$[\mathrm{KU}_2^\wedge, \mathrm{Cof}_{17}]_{\mathrm{KO}_2^\wedge} = \mathbf{Z}/32\{q\}, \quad qq = 17q, \quad [\mathrm{KU}_2^\wedge, \Sigma \mathrm{Cof}_{17}]_{\mathrm{KO}_2^\wedge} = 0,$$

Put $z = (\Sigma q)p$. Then $[\mathrm{Cof}_{17}, \Sigma \mathrm{Cof}_{17}]_{\mathrm{KO}_2^\wedge} = \mathbf{Z}/32\{z\}$.

Proof. The Galois equivalence $a \otimes b \mapsto (ab, ag(b))$ turns the complexification of $9 + 7g$ into the matrix F below. Over $\mathbf{Z}_2$ set

$$F = \begin{pmatrix} 9 & 7 \\ 7 & 9 \end{pmatrix}, \quad U = \begin{pmatrix} 1/9 & 0 \\ -7 & 9 \end{pmatrix}, \quad V = \begin{pmatrix} 1 & -7/9 \\ 0 & 1 \end{pmatrix}.$$

Then $UFV = \mathrm{diag}(1, 32)$, proving the equivalence. Let e_0, e_1 be the images of the standard basis in $\mathrm{coker} F$. Conjugation exchanges these classes, and $e_1 = -(9/7)e_0 = 17e_0$ modulo 32, which gives the character.

The Galois endomorphism equivalence $\mathrm{map}_{\mathrm{KO}_2^\wedge}(\mathrm{KU}_2^\wedge, \mathrm{KU}_2^\wedge) \simeq \mathrm{KU}_2^\wedge \langle C_2 \rangle$ [Rog08, Lemma 6.1.2(b)] identifies postcomposition by $9 + 7g$, in degree $2j$, with

$$\begin{pmatrix} 9 & 7(-1)^j \\ 7(-1)^j & 9 \end{pmatrix}.$$

Its determinant is 32, so it is injective. Applying the mapping spectrum to (5.4) gives the asserted odd vanishing and the cyclic degree-zero cokernel generated by q , with $qq = 17q$. Applying $[-, \Sigma \mathrm{Cof}_{17}]_{\mathrm{KO}_2^\wedge}$ gives

$$[\mathrm{KU}_2^\wedge, \mathrm{Cof}_{17}]_{\mathrm{KO}_2^\wedge} \xrightarrow{(9+7g)^*} [\mathrm{KU}_2^\wedge, \mathrm{Cof}_{17}]_{\mathrm{KO}_2^\wedge} \longrightarrow [\mathrm{Cof}_{17}, \Sigma \mathrm{Cof}_{17}]_{\mathrm{KO}_2^\wedge} \longrightarrow 0.$$

The first map kills the generator q and the second sends it to z . $\square$

Lemma 5.7. *If a $\mathrm{KO}_2^\wedge$ -module L has a $\mathrm{KU}_2^\wedge$ -linear equivalence $\mathrm{KU}_2^\wedge \otimes_{\mathrm{KO}_2^\wedge} L \simeq \mathrm{KU}_2^\wedge/32$ with degree-zero character 17, then $L \simeq \mathrm{Cof}_{17}$ as a $\mathrm{KO}_2^\wedge$ -module.*

Proof. Trace duality gives a natural equivalence

$$\mathrm{map}_{\mathrm{KO}_2^\wedge}(\mathrm{KU}_2^\wedge, L) \simeq \mathrm{KU}_2^\wedge \otimes_{\mathrm{KO}_2^\wedge} L$$

[Rog08, Proposition 6.4.7]. Precomposition by g corresponds to the Galois action: the trace pairing satisfies $\mathrm{tr}(ag(b)) = \mathrm{tr}(g(a)b)$. Choose a generator $\ell \in [\mathrm{KU}_2^\wedge, L]_{\mathrm{KO}_2^\wedge} = \mathbf{Z}/32$. Then

$$\ell(9 + 7g) = (9 + 7 \cdot 17)\ell = 128\ell = 0.$$

It extends to $h: \mathrm{Cof}_{17} \rightarrow L$ with $hq = \ell$. Applying $\mathrm{map}_{\mathrm{KO}_2^\wedge}(\mathrm{KU}_2^\wedge, -)$ gives a map between modules equivalent to $\mathrm{KU}_2^\wedge/32$ which carries a degree-zero generator to a generator. Bott periodicity makes this an equivalence, and faithfulness of complexification makes h an equivalence. $\square$

Proposition 5.8. *Under any $KU_2^\wedge$ -linear trivialization of $KU_2^\wedge \otimes_{KO_2^\wedge} \text{Cof}_{17}$, the image of complexification is*

$$\begin{aligned} \text{im}([\text{Cof}_{17}, \Sigma \text{Cof}_{17}]_{KO_2^\wedge} &\longrightarrow [KU_2^\wedge/32, \Sigma(KU_2^\wedge/32)]_{KU_2^\wedge}) \\ &= 2(\mathbf{Z}/32)\{Q_{KU_2^\wedge/32}\}. \end{aligned}$$

Proof. Evenness and the cofiber sequence for 32 identify the target with $\mathbf{Z}/32\{Q_{KU_2^\wedge/32}\}$. It suffices to complexify z . Put $D = \text{diag}(1, 32)$ and let $\varphi: \text{cofib } F \rightarrow \text{cofib } D$ be induced by V^{-1} on sources and U on targets. For the respective cofiber maps, naturality gives $\varphi q_F = q_D U$ and $p_D \varphi = (\Sigma V^{-1}) p_F$. Consequently

$$(\Sigma \varphi)(\Sigma q_F) p_F \varphi^{-1} = (\Sigma q_D) \Sigma(UV) p_D.$$

The first diagonal summand has contractible cofiber. On the second,

$$(UV)_{22} = \frac{49}{9} + 9 = \frac{130}{9} \equiv 18 \pmod{32}.$$

Thus z complexifies to $18Q_{KU_2^\wedge/32}$, which generates the even ideal. The endomorphism calculation

$$\pi_0 \text{End}_{KU_2^\wedge}(KU_2^\wedge/32) = \mathbf{Z}/32$$

shows that changes of trivialization are odd scalars. They commute with the Bockstein, so the image is independent of that choice. $\square$

5.4. Connections and the exclusion of the line. Let R be a commutative ring spectrum, let A be an $\mathbb{E}_1$ - R -algebra, and let $d: A \rightarrow \Sigma A$ be a coherent R -relative derivation. Write $s_d = (\text{id}, d)$ and $s_0 = (\text{id}, 0)$ for the algebra sections of $A \oplus \Sigma A \rightarrow A$. For a left A -module M define its jet module

$$J_d(M) = s_d^*((A \oplus \Sigma A) \otimes_{A, s_0} M).$$

A connection is an A -linear section of $J_d(M) \rightarrow M$. As an R -module $J_d(M) \simeq M \oplus \Sigma M$, so its underlying map is (id, ∇) for an R -linear map $\nabla: M \rightarrow \Sigma M$.

Lemma 5.9. *There is a natural fiber sequence of A -modules*

$$\Sigma M \longrightarrow J_d(M) \longrightarrow M \longrightarrow \Sigma^2 M.$$

A connection exists if $[M, \Sigma^2 M]_A = 0$, and its homotopy classes then form a torsor for $[M, \Sigma M]_A$. Base change of commutative ground rings preserves the sequence and its sections. On $M = A$, the section s_d is a connection; if $\pi_{-1}A = 0$, every connection on A has underlying R -linear homotopy class d .

Proof. Tensor the sequence of $(A \oplus \Sigma A)$ - A -bimodules $\Sigma A \rightarrow A \oplus \Sigma A \rightarrow A$ on the right with M , and restrict the left action along s_d . The right action on the middle term is through s_0 . The composite of s_d with projection is the identity, which gives the sequence. Its connecting class is the obstruction to splitting, and its sections, when nonempty, form a torsor under $\text{Map}_A(M, \Sigma M)$. Base change is exact and symmetric monoidal, so it preserves this construction. For $M = A$, the map s_d is A -linear for the restricted action and splits projection. The last assertion follows from $[A, \Sigma A]_A = \pi_{-1}A$. $\square$

Proposition 5.10. *There is no invertible $KO_2^\wedge/32$ -module whose complexification is free and whose degree-zero character is 17.*

Proof. For such a line L , tensor cancellation gives

$$[L, \Sigma^2 L]_{KO_2^\wedge/32} = \pi_{-2}(KO_2^\wedge/32) = 0,$$

the last equality following from $\pi_{-2}KO_2^\wedge = \pi_{-3}KO_2^\wedge = 0$. Proposition 5.5 and Lemma 5.9 give a connection $\nabla: L \rightarrow \Sigma L$ in $\text{Mod}_{KO_2^\wedge}$. After complexification and a $KU_2^\wedge/32$ -linear trivialization it is a connection on the free module for $Q_{KU_2^\wedge/32}$. Since $\pi_{-1}(KU_2^\wedge/32) = 0$, its underlying homotopy class is exactly that Bockstein. Lemma 5.7 identifies the underlying real module with Cof_{17} , contradicting Proposition 5.8. $\square$

Proof of the group assertion in Theorem 5.1. Proposition 5.10 and Lemma 5.2 show that suspension generates. The real Bott unit gives period eight, whereas $\pi_1(\mathrm{KO}_2^\wedge/32) = \mathbf{Z}/2$ and $\pi_5(\mathrm{KO}_2^\wedge/32) = 0$ rule out period four. The suspension therefore has order exactly eight, resolving the extension. $\square$

5.5. The differential on the character 17.

Lemma 5.11. *The fixed $\mathbb{E}_1$ -structure underlying $\mathrm{KO}_2^\wedge/32$ admits an $\mathbb{E}_1$ - $\mathrm{KO}_2^\wedge$ -algebra map to $\mathrm{KO}_2^\wedge/16$ whose underlying map is reduction, for a synthetic quotient $\mathbb{E}_1$ -structure on the target.*

Proof. Burklund's construction gives an $\mathbb{E}_1$ -structure on $\nu\mathbb{S}_2^\wedge/\tilde{2}^4$ [Bur22, Theorem 3.2]. The cofiber square supplies a unit-preserving underlying reduction map. Lemma 5.3 places its arity- r obstruction in

$$[\Sigma^{r-2}(S^{1,4})^{\otimes r}, \nu\mathbb{S}_2^\wedge/\tilde{2}^4] = \pi_{2r-2, 3r+2}(\nu\mathbb{S}_2^\wedge/\tilde{2}^4).$$

This vanishes by [Bur22, Lemma 3.3], since $3r + 2 > \frac{1}{2}(2r - 2) + 4$ for every $r \geq 2$. Choose the coherences, invert τ , and base change to $\mathrm{KO}_2^\wedge$. $\square$

Proof of (5.1). Suppose $d_3(\chi_{17}) = 0$. For $j = 4, 5$, the component of the free module is

$$B \mathrm{Aut}_{\mathrm{KU}_2^\wedge/2^j}(\mathrm{KU}_2^\wedge/2^j) \simeq B \mathrm{GL}_1(\mathrm{KU}_2^\wedge/2^j).$$

Lemma 5.11 gives a based equivariant map $B \mathrm{GL}_1(\mathrm{KU}_2^\wedge/32) \rightarrow B \mathrm{GL}_1(\mathrm{KU}_2^\wedge/16)$. These spaces require only associative quotient structures. Their fundamental groups are the scalar units, which act trivially on higher homotopy. Since $\pi_2 B \mathrm{GL}_1(\mathrm{KU}_2^\wedge/32) = \pi_4 B \mathrm{GL}_1(\mathrm{KU}_2^\wedge/32) = 0$, our assumption gives partial descent data through the fourth Postnikov stage. Its next obstruction is

$$o_5 \in H^6(C_2; \pi_4(\mathrm{KU}_2^\wedge/32)).$$

Reduction makes the character trivial because $17 \equiv 1 \pmod{16}$. With an identification at the first Postnikov stage fixed, the reduced partial data form a torsor for $H^3(C_2; (\mathbf{Z}/16)u) = \mathbf{Z}/2$. Both classes extend to full data: they are given by $\mathrm{KO}_2^\wedge/16$ and $\Sigma^4(\mathrm{KO}_2^\wedge/16)$, the latter trivialized after complexification by u^2 . Their difference is nonzero because the filtration-three suspension class in Lemma 5.2 maps nontrivially under

$$H^3(C_2; \mathbf{Z}_2 u) \xrightarrow{\sim} H^3(C_2; (\mathbf{Z}/16)u).$$

The reduced obstruction is therefore zero. But its coefficient map

$$H^6(C_2; (\mathbf{Z}/32)u^2) \xrightarrow{\sim} H^6(C_2; (\mathbf{Z}/16)u^2)$$

is an isomorphism: for trivial coefficients, positive even cohomology is the quotient by two. Thus $o_5 = 0$. Proposition 5.10 excludes survival, and Lemma 5.2 leaves only d_3 and d_5 . Hence $d_3(\chi_{17})$ is nonzero. Its target has unique nonzero element $[16u]$, giving the formula. $\square$

6. FINITE DOUBLE QUOTIENTS

Fix $m \geq 5$ and the Moore multiplication stipulated in Theorem 1.3. For $m \geq 6$ we use the compatible structures in [LLZ26, Sections 2.3 and 3.3]; for $m = 5$ we use Section 5. No map between different values of m is used in this section. The eta-stage maps are induced by the commutative tower A_n , and $\mathrm{KO}/(2^m, \eta^n) = A_n \otimes (\mathbb{S}/2^m)$ has its resulting $\mathbb{E}_2$ structure. Its module category is monoidal, so its Picard space is a grouplike $\mathbb{E}_1$ -space.

The coefficient inputs are

$$(6.1) \quad \mathrm{Pic}(\mathrm{KU}/2^m) = \mathbf{Z}/2, \quad \mathrm{Pic}(\mathrm{KO}/2^m) = \mathbf{Z}/8, \quad d_3[1 + 2^{m-1}] \neq 0.$$

The complex assertion is [LLZ26, Theorem 2.25 and Example 2.27]. For the real assertions use [LLZ26, Theorem 3.23 and Proposition 3.26] when $m \geq 6$, and Theorem 5.1 when $m = 5$. All these Picard groups are generated by suspension.

By Proposition 2.3, base change to $KU/2^m$ gives cochains on the antipodal sphere $S(n\sigma)$. Equivariant local systems on this free sphere are sections over $\mathbf{RP}^{n-1}$ of the coefficient Picard space with conjugation monodromy. For a nonnegative skeleton dimension N , put

$$\text{PicSec}_{KU/2^m}(N) = \pi_0 \Gamma(\mathbf{RP}^N; \mathcal{P}(KU/2^m)).$$

For the n th eta stage take $N = n - 1$. Lemma 4.2 and Proposition 4.3 identify $\text{Pic}(KO/(2^m, \eta^n))$ with the subgroup whose pullback to S^{n-1} is realizable by cochain modules. For a higher odd sphere this means an even clutching coefficient. We first compute the entire section group, including its extensions, and then impose this condition.

Proposition 6.1. *For $n \geq 11$, with $N = n - 1$, there is a group isomorphism*

$$\text{PicSec}_{KU/2^m}(N) \cong \mathbf{Z}/8 \oplus \pi_{-n-2}(KO/2^m).$$

The first factor is suspension and the second is the kernel of restriction to $\mathbf{RP}^4$. On that kernel, pullback to the sphere is complexification followed by the Bott identification of degrees.

Lemma 6.2. *In a stable monoidal category with tensor product exact in both variables, the suspension of the unit is central in the Picard group. This also holds for the section groups above.*

Proof. The natural equivalences $(\Sigma \mathbf{1}) \otimes M \simeq \Sigma M \simeq M \otimes (\Sigma \mathbf{1})$ are preserved by exact monoidal functors, including transport and sections. Thus a split suspension subgroup acts trivially on its complementary kernel. $\square$

6.1. Additive coefficients and finite Postnikov intervals. The spaces $\mathcal{P}(KU/2^m)$ and $\Omega^\infty \Sigma(KU/2^m)$, the latter with its additive structure, have the same homotopy groups in degrees at least two. Their fundamental groups are $(\mathbf{Z}/2^m)^\times$ and $\mathbf{Z}/2^m$, respectively.

Lemma 6.3. *For $a \geq 3$ there is a natural equivalence of grouplike $\mathbb{E}_1$ -spaces, compatible with conjugation,*

$$(6.2) \quad \tau_{[a, 2a-2]} \mathcal{P}(KU/2^m) \simeq \Omega^\infty \tau_{[a, 2a-2]} \Sigma(KU/2^m).$$

Consequently the Picard d_3 with internal degree $t \geq 4$ agrees with the shifted additive differential. The differentials out of positive total degree with $s > 0$ also agree whenever they are defined in the section obstruction sequence.

Proof. The identity component of $\Omega \mathcal{P}(KU/2^m) = \text{GL}_1(KU/2^m)$, translated by the unit, is the zero component of $\Omega^\infty(KU/2^m)$ as a pointed space with action. The stable-range theorem [MS16, Theorem 5.1.2] identifies pointed spaces with spectra in $[a, 2a - 2]$, and the infinite loop functor is fully faithful in $[a - 1, 2a - 3]$. Hence this loop-space identification uniquely deloops in the stated interval, by the same argument as [MS16, Corollary 5.2.3]. It preserves the $\mathbb{E}_1$ structures: finite products remain in the fully faithful range, and their lifts are biproducts of spectra, with the unique unital additive multiplication and its coherences. Functoriality also preserves the coherent action.

For d_3 retain $[t, t + 2]$; even-row vanishing makes $d_2 = 0$, so source and target are unchanged by this truncation. For positive total degree and $s > 0$, loop the section problem once. Its identity-component problem for GL_1 is the additive pointed-space problem just described. The Postnikov obstructions therefore agree. This is the space-level argument of [MS16, Comparison Tool 5.2.4]. $\square$

Lemma 6.4. *For $m \geq 2$, the coefficients of $KO/2^m$ and complexification, up to eight-periodicity, are*

$d \bmod 8$	0	1	2	3	4	5	6	7
$\pi_d(KO/2^m)$	$(\mathbf{Z}/2^m)$	$\mathbf{Z}/2$	$(\mathbf{Z}/2)^2$	$\mathbf{Z}/2$	$(\mathbf{Z}/2^m)$	0	0	0
c	1	0	$(0, 2^{m-1}u)$	0	2	0	0	0

In degree two the generators may be chosen as the image of η^2 and a class with Bockstein η .

Proof. The coefficient exact sequence gives all entries except the degree-two extension. Lift η to $\pi_2(\mathrm{KO}/2)$. The map to $\mathrm{KO}/2^m$ induced by the identity on the source copy of KO and multiplication by 2^{m-1} on the target carries this lift to b_m with Bockstein η . Twice the original lift is an image of η^2 , killed by 2^{m-1} ; hence $2b_m = 0$. The image of η^2 is nonzero and has Bockstein zero, which splits the extension. The Wood sequence tensored with $\mathbb{S}/2^m$ shows that the image of $\pi_2(\mathrm{KO}/2^m) \rightarrow \pi_2(\mathrm{KU}/2^m)$ is the subgroup of order two: the following boundary has image $\ker(\eta: (\mathbf{Z}/2^m) \rightarrow \mathbf{Z}/2) = 2(\mathbf{Z}/2^m)$. Since η^2 complexifies to zero, $c(b_m) = 2^{m-1}u$. The degree-zero and degree-four assertions are the usual complexification formulas [Rog08, Section 5.3]. $\square$

Let h be the reduction of the integral eta class in bidegree $(1, 2)$, and let $z \in H^0(C_2; \pi_2(\mathrm{KU}/2^m))$ be the invariant class represented by $2^{m-1}u$.

Lemma 6.5. *The additive C_2 -homotopy fixed point sequence for $\mathrm{KU}/2^m$ has*

$$E_2 = (\mathbf{Z}/2^m)[u^{\pm 2}, h, z]/(2h, 2z, z^2), \quad |u^2| = (0, 4), \quad |h| = (1, 2), \quad |z| = (0, 2),$$

and $d_3(u^2) = h^3$, $d_3h = d_3z = 0$. Thus, for every integer j ,

$$d_3(h^s u^{2j}) = j h^{s+3} u^{2j-2}, \quad d_3(z h^s u^{2j}) = j z h^{s+3} u^{2j-2}.$$

Proof. The two-periodic resolution for C_2 with action $(-1)^q$ on $(\mathbf{Z}/2^m)u^q$ gives the groups and products. Reduction preserves the nonzero target of the integral differential $d_3(u^2) = h^3$ [HS14, Section 5]. The class h is detected by η , and z is detected by Lemma 6.4; both are permanent. Multiplicativity gives the formulas, including negative powers of u^2 . $\square$

For $Y = \mathcal{P}(\mathrm{KU}/2^m)$ or $\Omega^\infty \Sigma(\mathrm{KU}/2^m)$, the finite section sequence is

$$(6.3) \quad E_2^{s,t} = H^s(\mathbf{RP}^N; \pi_t Y) \implies \pi_{t-s} \Gamma(\mathbf{RP}^N; Y), \quad 0 \leq s \leq N.$$

Only the Postnikov section through degree N affects π_0 , so its filtration is finite. We use the grouplike $\mathbb{E}_1$ structure on $\mathcal{P}(\mathrm{KU}/2^m)$, without a global Picard spectrum for the finite quotient.

Lemma 6.6. *For $N \geq 5$, all zero-diagonal terms of (6.3) with $5 \leq s = t \leq N - 3$ vanish on E_4 for $Y = \mathcal{P}(\mathrm{KU}/2^m)$. For $Y = \Omega^\infty \Sigma(\mathrm{KU}/2^m)$ and $N \geq 6$, this holds for every $s = t \leq N - 3$. On the total-degree-one diagonal for $\tau_{\leq q} Y$, where $q \leq N$, every positive-degree term with $t \leq q - 2$ vanishes on E_4 , except possibly the scalar term $(s, t) = (0, 1)$.*

Proof. Label the coefficients by the additive monomials in Lemma 6.5, with internal degree shifted by one. For positive cohomological degree below the top column, every coefficient is $\mathbf{F}_2$. A monomial with odd exponent of u^2 supports a nonzero d_3 ; one with even exponent is hit by the monomial with exponent one higher, three columns to its left and two internal degrees lower. For the asserted Picard terms both source and target are retained, and Lemma 6.3 applies. If the target is the top column, the map $\mathbf{F}_2 \rightarrow (\mathbf{Z}/2^m)$ is the inclusion of $(\mathbf{Z}/2^m)[2]$, still injective. In the additive case the additional low terms $(1, 1)$ and $(3, 3)$ have labels zhu^{-2} and h^3u^{-2} ; their nonzero targets are $(4, 3)$ and $(6, 5)$.

For total degree one, the first positive-cohomology source is $(2, 3)$, labelled zh^2u^{-2} , and it supports a nonzero d_3 by the positive-stem comparison in Lemma 6.3. The same odd/even argument handles the later terms. When $t \leq q - 2$, the outgoing target is retained in $\tau_{\leq q} Y$, as are the incoming sources. Scalar classes lift from $\mathrm{KO}/2^m$: units for $\mathcal{P}(\mathrm{KU}/2^m)$, and all of $(\mathbf{Z}/2^m)$ for $\Omega^\infty \Sigma(\mathrm{KU}/2^m)$. $\square$

6.2. A comparison that preserves extensions. Let $F^b \pi_0 \Gamma(\mathbf{RP}^N; Y)$ be the kernel of restriction to $\mathbf{RP}^{b-1}$. It is also the image of $\pi_0 \Gamma(\mathbf{RP}^N; \tau_{\geq b} Y)$. Indeed, on $\mathbf{RP}^{b-1}$, passage to $\tau_{\leq b-1} Y$ is a bijection on section components by connectivity. Restriction of sections of this truncated coefficient space from $\mathbf{RP}^N$ to $\mathbf{RP}^{b-1}$ is injective on components: a homotopy extends across the remaining cells because all relevant coefficient homotopy groups vanish. The Postnikov fiber sequence then gives the claimed image.

Lemma 6.7. *Let $N \geq 10$, $a = N - 4$, and $b = N - 2$. For $Y = \mathcal{P}(\mathrm{KU}/2^m)$ or $Y = \Omega^\infty \Sigma(\mathrm{KU}/2^m)$, naturally*

$$(6.4) \quad \begin{aligned} F^b \pi_0 \Gamma(\mathbf{RP}^N; Y) &\cong \mathrm{im}(\pi_0 \Gamma(\mathbf{RP}^N; \tau_{[b, N]} Y) \\ &\longrightarrow \pi_0 \Gamma(\mathbf{RP}^N; \tau_{[a, N]} Y)). \end{aligned}$$

Consequently

$$F^b \mathrm{PicSec}_{\mathrm{KU}/2^m}(N) \cong F^b \pi_0 \Gamma(\mathbf{RP}^N; \Omega^\infty \Sigma(\mathrm{KU}/2^m))$$

as groups, compatibly with pullback to the covering sphere and with the coefficient identifications there.

Proof. The Postnikov fiber sequence gives

$$\begin{aligned} \pi_1 \Gamma(\mathbf{RP}^N; \tau_{\leq b-1} Y) &\xrightarrow{\partial} \pi_0 \Gamma(\mathbf{RP}^N; \tau_{\geq b} Y) \\ &\longrightarrow F^b \pi_0 \Gamma(\mathbf{RP}^N; Y) \longrightarrow 0. \end{aligned}$$

We show that $\mathrm{im} \partial$ already comes from $\pi_1 \Gamma(\mathbf{RP}^N; \tau_{[a, b-1]} Y)$; this identifies the quotient using only the finite interval in (6.4).

Consider

$$\pi_1 \Gamma(\mathbf{RP}^N; \tau_{\leq b-1} Y) \longrightarrow \pi_1 \Gamma(\mathbf{RP}^N; \tau_{\leq a-1} Y).$$

In the source, every nonscalar total-degree-one term with $t \leq a - 1$ has vanished by Lemma 6.6, applied with $q = b - 1 = a + 1$. The scalar filtration-zero term lifts from a global section of Y . After subtracting such a scalar, a source class has filtration at least $a - 1$, while the target has no total-degree-one filtration above $a - 2$. The remainder therefore maps to zero. This uses finite filtrations, so no unseen higher-filtration image remains in the target.

Subtract a global scalar before evaluating ∂ . The remainder lifts from $\pi_1 \Gamma(\mathbf{RP}^N; \tau_{[a, b-1]} Y)$, and the subtracted class has zero boundary. This proves the assertion about $\mathrm{im} \partial$. The sequence

$$\tau_{\geq b} Y \longrightarrow \tau_{\geq a} Y \longrightarrow \tau_{[a, b-1]} Y$$

then identifies the cokernel with the image of $\pi_0 \Gamma(\mathbf{RP}^N; \tau_{\geq b} Y)$ in $\pi_0 \Gamma(\mathbf{RP}^N; \tau_{\geq a} Y)$. Degrees above N do not affect these component groups, proving (6.4).

Since $N \leq 2a - 2$, Lemma 6.3 identifies the entire interval $[a, N]$ for the two coefficient spaces. It preserves multiplication, action, and truncation maps, and hence the images as groups. All constructions commute with pullback. On S^N the comparison reduces to the degree- N coefficient identification. $\square$

6.3. The first ten stages.

Proposition 6.8. *The first five groups in Theorem 1.3 have the asserted values. For $6 \leq n \leq 10$ the groups are*

n	6	7	8	9	10
$\mathrm{Pic}(\mathrm{KO}/(2^m, \eta^n))$	$\mathbf{Z}/8 \oplus \mathbf{Z}/2^{m-1}$	$\mathbf{Z}/8$	$\mathbf{Z}/8$	$\mathbf{Z}/8$	$\mathbf{Z}/8 \oplus \mathbf{Z}/2^m$

For $n \geq 5$ in this range, restriction to stage five retracts onto the suspension subgroup.

Proof. The unit group is $(\mathbf{Z}/2^m)^\times = \{\pm 1\} \times \langle 5 \rangle$, where 5 has order 2^{m-2} . The first stage is (6.1). On $\mathbf{RP}^1$, the parity-zero subgroup is identified with $(\mathbf{Z}/2^m)^\times$. Under this identification, the Bott trivialization identifies twice the suspension class with the unit -1 . All circular local systems are realizable, so the second group is $\mathbf{Z}/4 \oplus \mathbf{Z}/2^{m-2}$.

For $N = 2$, the diagonal consists of $\mathbf{Z}/2$ in degree zero and $H^1(\mathbf{RP}^2; (\mathbf{Z}/2^m)^\times) = ((\mathbf{Z}/2^m)^\times)[2] = (\mathbf{Z}/2)^2$. All survive, and pullback to S^2 is constant. The filtration-one subgroup has exponent two and twice the suspension class corresponds to the unit -1 under this identification. Centrality of suspension gives $\mathbf{Z}/4 \oplus \mathbf{Z}/2$. For $N = 3$, the additional term is $H^3(\mathbf{RP}^3; (\mathbf{Z}/2^m)^{\mathrm{sign}}) = \mathbf{Z}/2$. Its only possible incoming differential is from scalar

units, which lift globally. The group has order sixteen. Suspension has order eight: the map from $KO/2^m$ is an isomorphism in degree four, since $\eta^4 = 0$ and $\pi_{-1}(KO/2^m) = 0$, and complexification there is multiplication by two. Thus no degree-four unit exists. The central suspension subgroup has cyclic quotient, so the group is abelian. It cannot be $\mathbf{Z}/16$, since the suspension has nonzero parity whereas every order-eight element of $\mathbf{Z}/16$ maps trivially to $\mathbf{Z}/2$. The group is $\mathbf{Z}/8 \oplus \mathbf{Z}/2$. Pullback to S^3 has sign-invariant clutching class, hence lies in $(\mathbf{Z}/2^m)[2] \subset 2(\mathbf{Z}/2^m)$; all sections are realizable.

For $N = 4$, the extra unit $1 + 2^{m-1}$ is obstructed by (6.1). The map from its infinite-skeleton target $H^4(C_2; (\mathbf{Z}/2^m)^{\text{sign}}) = (\mathbf{Z}/2^m)[2]$ to $H^4(\mathbf{RP}^4; (\mathbf{Z}/2^m)^{\text{sign}}) = (\mathbf{Z}/2^m)$ is the nonzero inclusion. Only the eight suspensions remain, and $\text{PicSec}_{KU/2^m}(4) = \text{Pic}(KO/(2^m, \eta^5)) = \mathbf{Z}/8$.

At $N = 5$, the new top group is $H^5(\mathbf{RP}^5; (\mathbf{Z}/2^m)) = (\mathbf{Z}/2^m)$. Its incoming d_3 from $(2, 3)$ has image $(\mathbf{Z}/2^m)[2]$ by Lemmas 6.3 and 6.5. The only other possible source is scalar units, whose differential is zero. Thus the top subgroup is the actual quotient $(\mathbf{Z}/2^m)/(\mathbf{Z}/2^m)[2] = \mathbf{Z}/2^{m-1}$. Restriction to $N = 4$ splits off suspension, and Lemma 6.2 makes this a direct product. Top-cell pullback is multiplication by two, whose kernel is precisely the subgroup just quotiented out. Every class is realizable.

For $N = 6, 7, 8$, the possible new terms at $s = 5, 7$ are hit from $(2, 3)$ and $(4, 5)$, respectively. Only suspension remains. At $N = 9$, the top group is $H^9(\mathbf{RP}^9; (\mathbf{Z}/2^m)) = (\mathbf{Z}/2^m)$ and its incoming d_3 is zero: the source has additive label zh^6u^{-4} . Higher incoming differentials have positive-total-degree sources; all nonscalar such sources have already died by d_3 , while scalars are permanent. Thus the top subgroup is $(\mathbf{Z}/2^m)$. Pullback to S^9 is twice the clutching generator, so it is realizable. Restriction to $N = 4$ again splits off suspension. This proves the table and the retractions. $\square$

6.4. All higher stages.

Proof of Proposition 6.1. The map to $\text{PicSec}_{KU/2^m}(4) = \mathbf{Z}/8$ has its suspension section from $\text{Pic}(KO/2^m)$. By Lemma 6.6, its kernel F^5 equals F^{N-2} . Lemma 6.7 identifies this group with $F^{N-2}\pi_0\Gamma(\mathbf{RP}^N; \Omega^\infty\Sigma(KU/2^m))$. The low additive diagonal terms at $s = 1, 3$ and all interior terms have vanished, so this is the whole component group. Additive descent and Proposition 2.3 give

$$\Gamma(\mathbf{RP}^N; \Omega^\infty\Sigma(KU/2^m)) \simeq \Omega^\infty\Sigma(KO/(2^m, \eta^n)).$$

The module cofiber sequence

$$\Sigma^n(KO/2^m) \xrightarrow{\eta^n} KO/2^m \longrightarrow KO/(2^m, \eta^n)$$

and $\pi_{-1}(KO/2^m) = \pi_{-2}(KO/2^m) = 0$ yield

$$(6.5) \quad \pi_{-1}(KO/(2^m, \eta^n)) \xrightarrow{\sim} \pi_{-n-2}(KO/2^m).$$

Centrality of suspension gives the direct-product decomposition.

The interval comparison is natural under pullback to S^N and identifies $\pi_N\mathcal{P}(KU/2^m)$ with $\pi_{N-1}(KU/2^m)$. Complexification commutes with (6.5). The spherical-cochain and complexified cofiber splittings differ by the Bott equivalence, their degrees differing by $2n$. Thus reduced pullback is the stated complexification map followed by a Bott unit. In particular the condition of being even is unaffected. $\square$

Proof of Theorem 1.3. Proposition 6.8 handles $n \leq 10$. For $n \geq 11$, restrict Proposition 6.1 to realizable systems using Proposition 4.3. Suspension has zero reduced clutching class. If n is odd the sphere is even-dimensional and all sections are realizable. If n is even, the clutching coefficient must be even. In both cases the complementary subgroup is

$$\ker(\pi_{-n-2}(KO/2^m) \xrightarrow{c} \pi_{-n-2}(KU/2^m)/2).$$

By Lemma 6.4, in degree 0 modulo eight this kernel is $2(\mathbf{Z}/2^m) \cong \mathbf{Z}/2^{m-1}$; in degree four it is $(\mathbf{Z}/2^m)$; in degree two it is all of $(\mathbf{Z}/2)^2$, since $2^{m-1}u$ is even. In degrees one and three it is $\mathbf{Z}/2$, and all other entries vanish. Substitution of $d = -n - 2$ gives the residue table. The small-stage calculation gives the same formula for $6 \leq n \leq 10$. The splitting is defined by restriction to stage five, and Lemma 4.1 identifies the ordinary and local module categories for these finite quotients. $\square$

7. A RATIONAL OBSTRUCTION TO LIFTING THROUGH A QUOTIENT TOWER

Theorem 1.4 concerns an arbitrary prescribed transition in a tower, with the rational ring allowed to depend on that transition. Its proof has three parts. A punctured hypersurface realizes the quotient tower geometrically; a compact-unit comparison identifies its invertible modules with shifted line bundles; and an explicit Čech class obstructs extension of one line bundle. These assertions also give injectivity and the infinite-order cokernel class.

7.1. The punctured hypersurface. Fix $m \geq 1$, set $f = xy - zw$, and define

$$\text{Quad}_m = \mathbf{Q}[x, y, z, w, t]/(f - t^m), \quad X = \text{Spec}(\text{Quad}_m) \setminus \{(x, y, z, w, t)\}.$$

Regard

$$R_m = R\Gamma(X, \mathcal{O}_X)$$

as a commutative $H\mathbf{Q}$ -algebra, retaining its entire derived structure. The coordinate t gives a map $H(\mathbf{Q}[t]) \rightarrow R_m$. For $j \geq 1$ put

$$(7.1) \quad R_m/t^j = R_m \otimes_{H(\mathbf{Q}[t])} H(\mathbf{Q}[t]/(t^j)), \quad X_j = X \times_{\text{Spec}(\mathbf{Q}[t])} \text{Spec}(\mathbf{Q}[t]/(t^j)).$$

The quotient maps in $\mathbf{Q}[t]$ give compatible $\mathbb{E}_\infty$ -algebra maps $R_m/t^{j+1} \rightarrow R_m/t^j$.

These quotients refine the $\mathbb{E}_1$ descent tower of [LLZ26, Definition 2.10]. Indeed, consider the evaluation map $H(\mathbf{Q}[t]) \rightarrow H\mathbf{Q}$ at zero. Its recursive descent pullbacks can be formed in commutative $H(\mathbf{Q}[t])$ -algebras. By [LLZ26, Lemma 2.11], stage $j - 1$ has underlying unital module the cofiber of multiplication by t^j . It is discrete, its zeroth homotopy ring is $\mathbf{Q}[t]/(t^j)$, and surjectivity from $\mathbf{Q}[t]$ forces the quotient multiplication. This identifies the stages with $H(\mathbf{Q}[t]/(t^j))$, compatibly with their maps. Base change along $H(\mathbf{Q}[t]) \rightarrow R_m$ is exact on modules and preserves relative tensor products and finite pullbacks of algebras. It therefore identifies the resulting descent tower with (7.1). The underlying $\mathbb{E}_1$ quotient structures agree with those of Burklund's construction by [LLZ26, Lemma 2.13].

Lemma 7.1. *There are natural equivalences of $\mathbb{E}_\infty$ -rings*

$$R_m/t^j \simeq R\Gamma(X_j, \mathcal{O}_{X_j}),$$

compatible with the transition maps. The underlying R_m -module of R_m/t^j is the cofiber of multiplication by t^j .

Proof. The ring Quad_m is free over $\mathbf{Q}[x, y, z, w]$ on $1, t, \dots, t^{m-1}$. Multiplication by f is injective on this free module. Since $t^m = f$, multiplication by t , and hence by t^j , is injective. For the closed immersion $i_j : X_j \hookrightarrow X$, there is consequently an exact sequence

$$0 \longrightarrow \mathcal{O}_X \xrightarrow{t^j} \mathcal{O}_X \longrightarrow (i_j)_* \mathcal{O}_{X_j} \longrightarrow 0.$$

Applying derived global sections identifies the last term with the asserted cofiber. The two-term free resolution of $\mathbf{Q}[t]/(t^j)$ identifies the underlying module of (7.1) with the same cofiber. The canonical base-change map of commutative algebras realizes this identification, so it is an equivalence. The map and the resolutions are natural in the polynomial quotient maps. $\square$

Write

$$U = \operatorname{Spec}(\mathbf{Q}[x, y, z, w]/(xy - zw)) \setminus \{(x, y, z, w)\}.$$

Every X_j has underlying topological space U , and

$$(7.2) \quad X_m \cong U \times \operatorname{Spec}(\mathbf{Q}[t]/(t^m)).$$

The next neighborhood instead has equation $xy - zw = t^m$ modulo t^{m+1} .

Lemma 7.2. *Extension of scalars to the structure sheaf gives a symmetric monoidal equivalence*

$$\operatorname{Mod}_{R_m/t^j} \simeq D_{\text{qc}}(X_j).$$

It identifies Picard groups and transition maps as

$$\operatorname{Pic}(R_m/t^j) \cong \mathbf{Z} \oplus \operatorname{Pic}(X_j),$$

with the identity on the shift factor and restriction on line bundles.

Proof. For the quasi-compact open immersion $a_j : X_j \hookrightarrow \operatorname{Spec}(\operatorname{Quad}_m/(t^j))$, derived direct image preserves quasi-coherent cohomology. If $R\Gamma(X_j, M) = 0$, affine derived global sections give $R(a_j)_*M = 0$, and restriction then gives $M = 0$. Thus the structure sheaf generates $D_{\text{qc}}(X_j)$; see also [The26, Tag 0BQT]. It is compact because X_j is quasi-compact and separated [The26, Tag 09M1]. Here this can also be seen from the finite affine cover $D(x), D(z), D(w), D(y)$: its bounded Čech complex computes derived global sections and commutes with direct sums.

Compact-unit symmetric monoidal Morita theory [Mat16, Theorem 2.29], together with Lemma 7.1, gives the equivalence. Its left adjoint sends N to $\mathcal{O}_{X_j} \otimes_{R_m/t^j} N$, a formula compatible with restriction along $X_j \hookrightarrow X_{j+1}$.

An invertible complex on a scheme is locally a line bundle in one degree, with locally constant degree [The26, Tag 0FPY]. The polynomial $xy - zw$ is irreducible, so U and every X_j are connected. Invertible complexes are therefore precisely integer shifts of line bundles. This proves the Picard-group statement and its naturality. $\square$

7.2. The obstruction class. Order the standard cover of U by

$$U_1 = D(x), \quad U_2 = D(z), \quad U_3 = D(w), \quad U_4 = D(y),$$

and write $U_{i_0 \dots i_q}$ for intersections. Define transition functions by

$$(7.3) \quad g_{12} = g_{34} = 1, \quad g_{13} = g_{23} = \frac{w}{x}, \quad g_{14} = g_{24} = \frac{y}{z}.$$

These fractions are units on the indicated intersections. For example, $zw = xy$ makes x, y invertible on U_{23} and z, w invertible on U_{14} . With the convention $g_{ij}g_{jk}g_{ik}^{-1} = 1$ for $i < j < k$, the cocycle identities either hold identically or reduce to $zw/(xy) = 1$. Let L be this line bundle on U , and let L_m be its pullback under (7.2).

The square-zero ideal of $X_m \hookrightarrow X_{m+1}$ is

$$t^m \mathcal{O}_X / t^{m+1} \mathcal{O}_X \cong \mathcal{O}_U, \quad t^m a \longmapsto a \bmod t.$$

Regularity of t gives the identification, and $2m \geq m + 1$ gives square-zero multiplication. The unit-sheaf sequence therefore supplies

$$(7.4) \quad H^1(U, \mathcal{O}_U) \longrightarrow \operatorname{Pic}(X_{m+1}) \longrightarrow \operatorname{Pic}(X_m) \xrightarrow{\delta} H^2(U, \mathcal{O}_U);$$

see [The26, Tag 0C6R]. The line bundle L_m extends exactly when $\delta(L_m) = 0$.

The fractions in (7.3) also give units on the corresponding opens of X_{m+1} , since they are units modulo the nilpotent ideal (t) . Their multiplicative coboundary is 1 on 123 and 124, and

$$\frac{zw}{xy} = 1 - \frac{t^m}{xy}$$

on the other two triples. Thus $\delta(L_m)$ is represented on U by

$$(7.5) \quad (c_{123}, c_{124}, c_{134}, c_{234}) = \left(0, 0, -\frac{1}{xy}, -\frac{1}{zw}\right).$$

Lemma 7.3. *The class $[c] = \delta(L_m)$ is nonzero, whereas $H^1(U, \mathcal{O}_U) = 0$.*

Proof. Let $W = \text{Spec}(\mathbf{Q}[x, y, z, w]) \setminus \{(x, y, z, w)\}$, and let $b : U \hookrightarrow W$ be cut out by f . The exact sequence

$$0 \longrightarrow \mathcal{O}_W \xrightarrow{f} \mathcal{O}_W \longrightarrow b_*\mathcal{O}_U \longrightarrow 0$$

gives a connecting map $\partial : H^2(U, \mathcal{O}_U) \rightarrow H^3(W, \mathcal{O}_W)$. Lift (7.5) by the same fractions on the ordered cover $D(x), D(z), D(w), D(y)$ of W . Its additive differential is

$$d\tilde{c} = \tilde{c}_{234} - \tilde{c}_{134} + \tilde{c}_{124} - \tilde{c}_{123} = -\frac{1}{zw} + \frac{1}{xy} = -\frac{f}{xyzw}.$$

Writing $P = \mathbf{Q}[x, y, z, w]$ within this calculation, we obtain

$$(7.6) \quad \partial[c] = \left[-\frac{1}{xyzw}\right] \in H^3(W, \mathcal{O}_W) = \frac{P_{xyzw}}{P_{xyz} + P_{xyw} + P_{xzw} + P_{yzw}}.$$

The Laurent monomial $x^{-1}y^{-1}z^{-1}w^{-1}$ cannot lie in the denominator: every monomial in any of its four summands has a nonnegative exponent in at least one variable. Linear independence of Laurent monomials proves that (7.6) is nonzero.

The same Čech complex gives $H^1(W, \mathcal{O}_W) = H^2(W, \mathcal{O}_W) = 0$. Indeed, decompose it by Laurent monomials. A monomial whose negative-exponent set is N occurs exactly in summands whose inverted-coordinate set contains N . For nonempty proper N , adjoining a coordinate outside N gives a contracting homotopy. For N empty only degree zero remains, and for N all four coordinates only degree three remains. The long exact sequence above now gives $H^1(U, \mathcal{O}_U) = 0$. $\square$

Proof of Theorem 1.4. Lemmas 7.3 and 7.2, applied to (7.4), show that $\text{Pic}(R_m/t^{m+1}) \rightarrow \text{Pic}(R_m/t^m)$ is injective and excludes the class represented by L_m . A shift cannot change this obstruction, since restriction preserves the shift degree.

For every nonzero integer r , additivity of the connecting map gives

$$\delta(L_m^{\otimes r}) = r[c] \neq 0,$$

because $H^2(U, \mathcal{O}_U)$ is a $\mathbf{Q}$ -vector space. The cokernel class has infinite order. Finally,

$$\pi_{-1}(R_m/t) = H^1(U, \mathcal{O}_U) = 0$$

by Lemmas 7.1 and 7.3. $\square$

8. A HEIGHT-TWO OBSTRUCTION TO LIFTING FROM THE LOCAL SPHERE

Fix a prime $p \geq 5$. Let E be height-two Honda Morava E -theory. Use u_1 for its deformation coordinate and $u_E \in \pi_{-2}E$ for a periodicity unit. Then

$$E_0 = W(\mathbf{F}_{p^2})[[u_1]], \quad E_* = E_0[u_E^{\pm 1}].$$

Let $\mathbb{S}_2$ be the automorphism group of the height-two Honda formal group, and put $\mathbb{G}_2 = \mathbb{S}_2 \rtimes \text{Gal}(\mathbf{F}_{p^2}/\mathbf{F}_p)$ for its extended stabilizer group. Power-series rings carry their maximal-ideal topology, and finite quotients carry the discrete topology. Continuous first cohomology means continuous crossed homomorphisms modulo continuous coboundaries.

The proof of Theorem 1.5 first constructs a finite generalized Moore $\mathbb{E}_2$ -algebra with a specified Lubin–Tate quotient. An invariant square-zero element in that quotient gives a cocycle that cannot lift to $E_0^\times$. Descent then realizes this class by an invertible module and shows that any module induced from an invertible $K(2)$ -local spectrum would have a lift.

8.1. The Moore algebra. Use Araki generators

$$BP_* = \mathbf{Z}_{(p)}[v_1, v_2, \dots], \quad |v_i| = 2(p^i - 1).$$

We use the polynomial, p -torsion-free description of the Hopf algebroid (BP_*, BP_*BP) , with v_1 primitive modulo p , as in [Rav26, Section 4.1 and Appendix A2.2].

Lemma 8.1. *In a commutative ring C , $x \equiv y \pmod{p}$ implies*

$$x^{p^r} \equiv y^{p^r} \pmod{p^{r+1}} \quad (r \geq 0).$$

Proof. If $x - y \in p^a C$ with $a \geq 1$, the binomial expansion gives $x^p - y^p \in p^{a+1}C$: the intermediate binomial coefficients are divisible by p , and the last term has valuation at least $ap \geq a + 1$. Induct on r . $\square$

Lemma 8.2. *Let $d = 2(p-1)p^3$. For $1 \leq r \leq 4$, the degree- d primitives in the comodule $BP_*/(p^r)$ are precisely*

$$\text{Prim}_d(BP_*/(p^r)) = (\mathbf{Z}/p^r)v_1^{p^3}.$$

Proof. Lemma 8.1 applied to the two units of the Hopf algebroid shows that $v_1^{p^3}$ is primitive modulo p^4 . Modulo p , the calculation

$$\text{Ext}_{(BP_*, BP_*BP)}^0(BP_*, BP_*/p) = \mathbf{F}_p[v_1]$$

is [Rav26, Theorem 4.3.2(ii), $n = 1$], so the degree- d primitives are $\mathbf{F}_p v_1^{p^3}$. Inductively, subtract a scalar multiple of $v_1^{p^3}$ from a primitive modulo p^r and write the result as py_1 . Equality of its coactions modulo p^r , together with p -torsion-freeness of the polynomial cooperation ring, shows that y_1 is primitive modulo p^{r-1} . The induction proves the assertion. $\square$

Lemma 8.3. *For any unital ring structure on $\mathbb{S}/p^4$, there is an element*

$$v \in \pi_{2(p-1)p^3}(\mathbb{S}/p^4)$$

whose BP-Hurewicz image is $v_1^{p^3}$ under the unit-induced identification $BP_(\mathbb{S}/p^4) = BP_*/(p^4)$.*

Proof. The classical Adams self-map of the mod- p^r Moore spectrum has degree $2(p-1)p^{r-1}$ and is an equivalence in $\text{KU}_{(p)}$ -homology; see [ACL23, Section 6, opening paragraph]. At $r = 4$, let

$$a : \Sigma^{2(p-1)p^3}(\mathbb{S}/p^4) \longrightarrow \mathbb{S}/p^4$$

be this map, and compose it with the suspended algebra unit to obtain v' . Its BP-Hurewicz image is primitive, so Lemma 8.2 makes it $\lambda v_1^{p^3}$ for $\lambda \in \mathbf{Z}/p^4$.

The scalar λ is a unit. The unit generates $\text{KU}_*(\mathbb{S}/p^4) = \text{KU}_*/p^4$, and a sends its suspended generator to a generator. For the multiplicative formal group of complex K -theory, with Bott class $u_{\text{KU}} \in \pi_2 \text{KU}_{(p)}$, one has

$$[p](x) = \frac{1 - (1 - u_{\text{KU}}x)^p}{u_{\text{KU}}} \equiv u_{\text{KU}}^{p-1} x^p \pmod{p}.$$

A strict change to a p -typical coordinate preserves this leading coefficient modulo p . Thus the Araki generator v_1 maps to a unit after accounting for its Bott degree; the p -typical orientation conventions are those of [Rav26, Section 4.1 and Appendix A2.2]. Compatibility of Hurewicz maps with $BP \rightarrow \text{KU}_{(p)}$ forces λ to be invertible. Multiplying v' by λ^{-1} gives v . A different unit normalization changes the argument only by a unit scalar. $\square$

Proposition 8.4. *There is a finite generalized Moore $\mathbb{E}_2$ -algebra*

$$M = (\mathbb{S}/p^4)/v^p, \quad BP_*M = BP_*/(p^4, v_1^{p^4}),$$

of type two, where $\mathbb{S}/p^4$ has a chosen $\mathbb{E}_3$ -structure and v is as in Lemma 8.3. For a compatible Lubin–Tate coordinate, put

$$\overline{E}_0 = E_0/(p^4, u_1^{p^4}).$$

Then

$$(8.1) \quad \pi_*(E \otimes M) = \overline{E}_0[u_E^{\pm 1}].$$

The map $E_0 \rightarrow \overline{E}_0$ is the quotient map and is $\mathbb{G}_2$ -equivariant.

Proof. Burklund’s theorem gives an $\mathbb{E}_3$ -structure on $\mathbb{S}/p^4$ [Bur22, Theorem 1.2]. Choose the even-degree class v from Lemma 8.3. Its quotient admits a unital multiplication: the obstruction $Q_1(v)$ is annihilated by 2, which is invertible here [Bur22, Theorem 1.4, footnote 4]. That theorem, with $m = n = 2$ and $q = p > 2$, equips

$$\mathrm{cofib}(\Sigma^{2(p-1)p^4} \mathbb{S}/p^4 \xrightarrow{v^p} \mathbb{S}/p^4)$$

with an $\mathbb{E}_2$ -algebra structure over $\mathbb{S}/p^4$. We fix this structure on M .

The unit map makes $BP_*(\mathbb{S}/p^4)$ the quotient ring $BP_*/(p^4)$: it is surjective on homotopy groups and hence forces the product. The defining map for M acts by the non-zero-divisor $v_1^{p^4}$. Its homology exact sequence gives the displayed BP -homology in even degrees. The two cofiber constructions make M finite.

To check its type, it is rationally trivial. For $j = 1, 2$, smashing the first cofiber sequence with $K(j)$ gives an even part $K(j)_*$ generated by the unit and an odd shifted copy. The image of v in the even part is a unit for $j = 1$ and zero for $j = 2$, by the standard BP orientations. Thus multiplication by v^p is an equivalence after smashing with $K(1)$ and is null after smashing with $K(2)$. Hence $K(1)_*M = 0$ and $K(2)_*M \neq 0$.

Choose the p -typical universal deformation F with

$$[p]_F(x) = px +_F u_1 x^p +_F x^{p^2},$$

as in [BDM⁺19, Section 2.1]. Its graded orientation sends v_1 to $u_1 u_E^{1-p}$. The two cofiber sequences give

$$\pi_*(E \otimes \mathbb{S}/p^4) = E_*/p^4, \quad \pi_*(E \otimes M) = E_*/(p^4, u_1^{p^4}),$$

since both successive elements are non-zero-divisors in their respective rings. The units make these ring identifications canonical once the coordinate is fixed. The action is through the E factor, so the homotopy quotient map is equivariant. The invariance of its kernel is also checked directly in Lemma 8.7. $\square$

Lemma 8.5. *Every ordinary module over $L_{K(2)}M$ is $K(2)$ -local. Its ordinary and local relative tensor products agree. Base change from $K(2)$ -local spectra is therefore*

$$X \longmapsto (L_{K(2)}M) \otimes X.$$

The same locality assertion holds for modules over $E \otimes M$.

Proof. Since M is finite, $(L_{K(2)}\mathbb{S}) \otimes M$ is local: finite duality turns maps into it from a $K(2)$ -acyclic Z into maps from $Z \otimes DM$ to $L_{K(2)}\mathbb{S}$. It is $K(2)$ -equivalent to M , so

$$L_{K(2)}M \simeq (L_{K(2)}\mathbb{S}) \otimes M.$$

The Bousfield-class calculation [HS99, Proposition 5.3] gives

$$\langle L_{K(2)}\mathbb{S} \rangle = \langle K(0) \vee K(1) \vee K(2) \rangle.$$

Smashing with the finite type-two spectrum M gives $\langle L_{K(2)}M \rangle = \langle K(2) \rangle$; the smash $K(2) \otimes M$ is a nonzero finite wedge of suspensions of the graded-field theory $K(2)$.

For an ordinary $(L_{K(2)}M)$ -module N and an acyclic Z , the free–forgetful adjunction gives

$$\mathrm{Map}_{\mathrm{Sp}}(Z, N) \simeq \mathrm{Map}_{\mathrm{Mod}_{L_{K(2)}M}}((L_{K(2)}M) \otimes Z, N) \simeq *.$$

This proves locality, including that of relative tensor products. The same Bousfield equality shows that smashing with $L_{K(2)}M$ sends $K(2)$ -equivalences to equivalences, giving the stated base-change formula. Finally $E \otimes M$ is local by finite duality. Monoidal localization factors the algebra map $M \rightarrow E \otimes M$ through $L_{K(2)}M$, so every $(E \otimes M)$ -module is a module over the latter algebra and is local. $\square$

8.2. A cocycle detected at a central scalar.

Lemma 8.6. *Let a topological group H act continuously on a commutative topological ring C , and let $z \in Z(H)$ act trivially on C . Evaluation defines a homomorphism*

$$\text{ev}_z : H_c^1(H, C^\times) \longrightarrow (C^H)^\times,$$

natural in equivariant ring maps on which z acts trivially.

Proof. For a crossed homomorphism use the convention $a(gh) = a(g)g(a(h))$. Commutativity of $gz = zg$ and the trivial action of z give

$$a(g)g(a(z)) = a(gz) = a(zg) = a(z)a(g).$$

Cancellation shows that $a(z)$ is invariant. A coboundary $g \mapsto b^{-1}g(b)$ takes value 1 at z . Thus evaluation descends to cohomology, and multiplicativity and naturality follow from the pointwise formulas. $\square$

For $g \in \mathbb{S}_2$, there are $t_0(g) \in E_0^\times$ and $t_1(g) \in E_0$ such that

$$(8.2) \quad g(u_1) = t_0(g)^{p-1}u_1 + t_0(g)^{-1}t_1(g)(p - p^p).$$

The residue of $t_0(g)$ lies in $\mathbf{F}_{p^2}^\times$. Galois acts on the Witt vectors by Frobenius and fixes u_1 ; see [BDM⁺19, Section 2.1, (2.1.1), and Theorem 3.3(a)]. We also use

$$(8.3) \quad E_0^{\mathbb{G}_2} = \mathbf{Z}_p, \quad \mathbf{Z}_p^\times \subseteq Z(\mathbb{G}_2) \text{ acts trivially on } E_0.$$

Both facts hold at arbitrary height and prime [BG18, Lemma 1.33 and its proof]. For $a \in \mathbf{Z}_p^\times$, write $[a]$ for the resulting central scalar automorphism.

Lemma 8.7. *The ideal $(p^4, u_1^{p^4}) \subseteq E_0$ is $\mathbb{G}_2$ -invariant. In $\overline{E}_0$, the element*

$$\varepsilon = p^3 u_1^{p^4-1}$$

satisfies

$$\varepsilon \neq 0, \quad p\varepsilon = u_1\varepsilon = \varepsilon^2 = 0, \quad g(\varepsilon) = \varepsilon \quad (g \in \mathbb{G}_2).$$

Consequently $1 + \mathbf{F}_p\varepsilon$ is a trivially acted-on subgroup of $\overline{E}_0^\times$ of order p .

Proof. Equation (8.2) has the form $g(u_1) = au_1 + ph$ with a a unit. Lemma 8.1 gives $g(u_1)^{p^4} \equiv (au_1)^{p^4} \pmod{p^5}$, proving $g((p^4, u_1^{p^4})) \subseteq (p^4, u_1^{p^4})$. Apply the same argument to g^{-1} ; Galois invariance is immediate.

As a module over $W(\mathbf{F}_{p^2})/p^4$, the quotient $\overline{E}_0$ is free on $1, u_1, \dots, u_1^{p^4-1}$. This proves nonvanishing and the three annihilation assertions. Modulo p , equation (8.2) gives

$$g(\varepsilon) = \varepsilon t_0(g)^{(p-1)(p^4-1)}.$$

The exponent is divisible by $p^2 - 1$, since $p^4 \equiv 1 \pmod{p+1}$. Thus the factor on the right is 1 modulo (p, u_1) , an ideal that annihilates ε . Galois fixes p and u_1 , so ε is fixed by the full group. Finally, multiplication in $1 + \mathbf{F}_p\varepsilon$ adds coefficients, since $\varepsilon^2 = 0$. $\square$

Lemma 8.8. *There is a continuous homomorphism $\ell : \mathbb{G}_2 \rightarrow (\mathbf{F}_p, +)$ with $\ell([1+p]) = 1$.*

Proof. Let Frob denote the nontrivial automorphism of $W(\mathbf{F}_{p^2})$ over $\mathbf{Z}_p$. The Honda endomorphism order has the presentation

$$\mathcal{D} = W(\mathbf{F}_{p^2})\langle S \rangle / (S^2 = p, Sa = \text{Frob}(a)S), \quad \mathbb{S}_2 = \mathcal{D}^\times.$$

Galois acts by Frob on coefficients and fixes S [BDM⁺19, Section 2.1]. Every element is uniquely $a + bS$. The assignments

$$a \mapsto \begin{pmatrix} a & 0 \\ 0 & \text{Frob}(a) \end{pmatrix}, \quad S \mapsto \begin{pmatrix} 0 & p \\ 1 & 0 \end{pmatrix}$$

respect the relations and define a representation in $M_2(W(\mathbf{F}_{p^2}))$. Its determinant gives a continuous multiplicative map

$$\text{Nrd}(a + bS) = a\text{Frob}(a) - pb\text{Frob}(b) \in \mathbf{Z}_p.$$

On units it takes values in $\mathbf{Z}_p^\times$. The formula is Galois invariant, so it extends to $\mathbb{G}_2$ by assigning the Galois factor value 1. For a scalar $[a]$, its value is a^2 .

Define

$$(8.4) \quad \ell(g) = \frac{\text{Nrd}(g)^{p-1} - 1}{2p(p-1)} \pmod{p}.$$

The numerator is divisible by p , and $2(p-1)$ is a p -adic unit. If $a^{p-1} = 1 + pA$ and $b^{p-1} = 1 + pB$, then $(ab)^{p-1} \equiv 1 + p(A+B) \pmod{p^2}$. This proves additivity. The map factors through a finite quotient of $\mathbf{Z}_p^\times$, so it is continuous. Finally,

$$\text{Nrd}([1+p]) = (1+p)^2, \quad (1+p)^{2(p-1)} \equiv 1 + 2p(p-1) \pmod{p^2},$$

giving the required normalization. $\square$

Proposition 8.9. *The formula*

$$(8.5) \quad c(g) = 1 + \ell(g)\varepsilon = 1 + \ell(g)p^3u_1^{p^4-1}$$

defines a continuous crossed homomorphism $\mathbb{G}_2 \rightarrow \overline{E}_0^\times$. Its class has order p in $H_c^1(\mathbb{G}_2, \overline{E}_0^\times)$ and in the quotient by

$$\text{im}(H_c^1(\mathbb{G}_2, E_0^\times) \rightarrow H_c^1(\mathbb{G}_2, \overline{E}_0^\times)).$$

Proof. The value $\ell(g)\varepsilon$ is independent of a lift of $\ell(g)$ because $p\varepsilon = 0$. Continuity is immediate. Lemmas 8.7 and 8.8 give

$$c(g)g(c(h)) = (1 + \ell(g)\varepsilon)(1 + \ell(h)\varepsilon) = 1 + (\ell(g) + \ell(h))\varepsilon = c(gh),$$

and pointwise $c^p = 1$.

For any continuous crossed homomorphism $a : \mathbb{G}_2 \rightarrow E_0^\times$, Lemma 8.6 and (8.3) give $a([1+p]) \in \mathbf{Z}_p^\times$. Evaluation is unchanged by coboundaries, both before and after reduction. If $[c]^r$ lifted for some $1 \leq r < p$, its value

$$c([1+p])^r = 1 + r\varepsilon$$

would be the image of a scalar in $\mathbf{Z}_p^\times$. Setting $u_1 = 0$ in $\overline{E}_0$ forces that scalar to be 1 modulo p^4 , so its image would be 1. This contradicts $r\varepsilon \neq 0$. All the indicated powers are therefore outside the image. Together with $c^p = 1$, this proves both order assertions. $\square$

Corollary 8.10. *The connecting homomorphism for*

$$1 \rightarrow 1 + (p^4, u_1^{p^4}) \rightarrow E_0^\times \rightarrow \overline{E}_0^\times \rightarrow 1$$

satisfies $\delta[c] \neq 0$ in $H_c^2(\mathbb{G}_2, 1 + (p^4, u_1^{p^4}))$.

Proof. Reduction on units is surjective because $(p^4, u_1^{p^4})$ lies in the maximal ideal. It has a continuous set-theoretic section because its target is finite and discrete. The sequence therefore gives the usual long exact sequence of continuous cochains. Exactness and Proposition 8.9 prove the assertion. $\square$

8.3. Realization and base change. For the algebra of Proposition 8.4, the inequalities $2p - 1 > 4$ and $(p - 1) \nmid 2$ permit the Picard descent calculation of [LLZ26, Theorem 3.18]. It yields

$$(8.6) \quad 0 \longrightarrow H_c^1(\mathbb{G}_2, \overline{E}_0^\times) \xrightarrow{\iota} \text{Pic}(L_{K(2)}M) \xrightarrow{\text{par}} \mathbf{Z}/2 \longrightarrow 0.$$

Parity is detected after base change to $E \otimes M$. For the specified quotient structure, $\text{Pic}(E \otimes M) = \mathbf{Z}/2$ by [LLZ26, Example 2.27 and the paragraph following Theorem 3.1]. Lemma 8.5 identifies the ordinary module categories here with their local versions.

We fix the normalization of ι . If an invertible module L has parity zero, choose a degree-zero generator e_L of $\pi_*((E \otimes M) \otimes_{L_{K(2)}M} L)$. Its semilinear descent action has the form

$$g(e_L) = a_L(g)e_L, \quad a_L(g) \in \overline{E}_0^\times.$$

Replacing e_L by be_L changes the cocycle to $b^{-1}a_L(g)g(b)$. Thus its cohomology class is intrinsic. It is the degree-one descent invariant, and ι is its inverse on the parity kernel. Equivariant extension of scalars carries the generator and its automorphisms to their images, so this invariant is natural, with $E_0 \rightarrow \overline{E}_0$ acting by reduction of cocycles.

The group $\text{Pic}(L_{K(2)}M)$ is abelian. Its even subgroup in (8.6) is abelian, and every odd class is the product of an even class with suspension, which is central by Lemma 6.2.

Lemma 8.11. *If X is invertible in $\text{Sp}_{K(2)}$ and its base change $(L_{K(2)}M) \otimes X$ has even parity, then*

$$\iota^{-1}([(L_{K(2)}M) \otimes X]) \in \text{im}(H_c^1(\mathbb{G}_2, E_0^\times) \longrightarrow H_c^1(\mathbb{G}_2, \overline{E}_0^\times)).$$

Proof. The local E -module $E \widehat{\otimes} X = L_{K(2)}(E \otimes X)$ is invertible. Its Picard group is $\mathbf{Z}/2$, detected by suspension parity [LLZ26, Proposition 2.23 and Example 2.27]. Hence its homotopy is free of rank one over E_* in one parity. Base change to $E \otimes M$ preserves this parity because E_* and $\overline{E}_0[u_E^{\pm 1}]$ are even periodic and $\overline{E}_0 \neq 0$. The assumed parity is therefore even; choose a generator $e \in \pi_0(E \widehat{\otimes} X)$.

The completed Morava module carries a continuous semilinear $\mathbb{G}_2$ -action for its maximal-ideal topology [BSSW25, Section 2.2]. Since X is invertible and thus dualizable, this is its natural topology [HS99, Proposition 11.9]. The generator identifies the degree-zero module homeomorphically with E_0 . Thus

$$g(e) = a(g)e$$

defines a continuous crossed homomorphism $a : \mathbb{G}_2 \rightarrow E_0^\times$.

There is a natural equivariant equivalence

$$(E \otimes M) \otimes_E (E \widehat{\otimes} X) \simeq (E \otimes M) \otimes_{L_{K(2)}M} ((L_{K(2)}M) \otimes X).$$

After localization, both sides identify by extension of scalars with $(E \otimes M) \otimes X$; Lemma 8.5 says that their module spectra are already local. The image of e generates the homotopy on the right, and its cocycle is $a(g) \bmod (p^4, u_1^{p^4})$. The normalization of (8.6) proves the claim. $\square$

Proof of Theorem 1.5. Choose the finite type-two $\mathbb{E}_2$ -algebra of Proposition 8.4, and let $[L] = \iota([c])$ under (8.6). Proposition 8.9 and injectivity of ι show that $[L]$ has order p . If $[L]^r = [(L_{K(2)}M) \otimes X]$ for $1 \leq r < p$, this class has parity zero. Lemma 8.11 would force $[c]^r$ to lift to $H_c^1(\mathbb{G}_2, E_0^\times)$, contrary to Proposition 8.9. Every nontrivial power remains nontrivial in the cokernel of the base-change homomorphism, giving the required subgroup $\mathbf{Z}/p$. $\square$

Example 8.12. At $p = 5$, the construction has

$$M = (\mathbb{S}/625)/v^5, \quad \overline{E}_0 = W(\mathbf{F}_{25})[[u_1]]/(625, u_1^{625}),$$

and the obstructing cocycle is

$$c(g) = 1 + \ell(g)125u_1^{624}.$$

Its value at the central scalar [6] is $1 + 125u_1^{624}$. Thus the obstruction is detected without a computation of the full Picard group or the minimal v_2 -periodicity of M .

APPENDIX A. LOW-EXPONENT MULTIPLICATION STRUCTURES

The Moore multiplications in Theorem 1.3 specify the algebras whose Picard groups are computed. A multiplication on a complex quotient need not arise from a multiplication on the corresponding Moore spectrum.

Proposition A.1. *For $m = 3, 4$, give $KU_2^\wedge/2^m$ the regular-quotient $\mathbb{E}_1$ - $KU_2^\wedge$ -algebra structure of [LLZ26, Construction 2.20]. It admits an $\mathbb{E}_2$ refinement, and every such refinement has Picard group $\mathbf{Z}/2$, generated by suspension. The same holds for the integral finite quotients and their local module categories.*

Proof. The even-periodic coefficient ring $\mathbf{Z}_2[u^{\pm 1}]$ is complete regular local in the graded sense, with the regular sequence consisting of 2. For a sequence of length one, [LLZ26, Remark 2.26] provides the refinement when its exponent is strictly greater than two. Theorem 2.25 of that reference makes suspension generate the Picard group. Bott periodicity gives order at most two, and the nonzero even coefficients and zero odd coefficients exclude odd periodicity. Completion preserves these finite quotients, and Lemma 4.1 proves the last assertion. $\square$

Thus Proposition A.1 specifies the first eta stages $KO/(8, \eta)$ and $KO/(16, \eta)$. To formulate nonexistence without assuming a multiplication, consider, for $m = 1, 2$, the spectrum

$$A_n \otimes (\mathbb{S}/2^m)$$

as a pointed spectrum, with unit the composite of the structural and cofiber maps.

Theorem A.2. *For every $n \geq 1$, $A_n \otimes (\mathbb{S}/2)$ admits no $\mathbb{E}_2$ -ring structure with this specified unit.*

Proof. Use $KU/2$ as a right KO -module. Its KU -linear actions of η and 2 are null. The first follows from $\pi_1 KU = 0$; for the second, the cofiber sequence for 2 and evenness give $[KU/2, KU/2]_{KU} = \mathbf{F}_2$ by restriction to the unit. Proposition 2.3 consequently gives

$$(KU/2) \otimes_{KO} A_n \otimes (\mathbb{S}/2) \simeq KU/2 \oplus \Sigma(KU/2) \oplus \Sigma^{n+1}(KU/2) \oplus \Sigma^{n+2}(KU/2) \neq 0.$$

An $\mathbb{E}_2$ multiplication with the specified unit would satisfy $2 \cdot 1 = 0$. The Hopkins–Mahowald theorem in its universal-property form would give a unital $\mathbb{E}_2$ map $H\mathbf{F}_2 \rightarrow A_n \otimes (\mathbb{S}/2)$ [ACB19, Theorem 5.1]. Since $KU/2$ has the Bousfield class of $K(1)$ and $(KU/2) \otimes H\mathbf{F}_2 = 0$, this would imply $(KU/2) \otimes A_n \otimes (\mathbb{S}/2) = 0$; see [ACB19, Corollary 5.4]. Every term in the two-sided bar construction of $(KU/2) \otimes_{KO} A_n \otimes (\mathbb{S}/2)$ would then vanish, contradicting the displayed equivalence. $\square$

Lemma A.3. *Let $e: \mathbb{S}/4 \rightarrow KO/4$ be induced by the unit. For every finite spectrum Z with cells in degrees 0, 1, 2, the map*

$$e_*: [Z, \mathbb{S}/4] \xrightarrow{\sim} [Z, KO/4]$$

is an isomorphism. This applies to $Z = \mathbb{S}/4$ and $Z = (\mathbb{S}/4) \otimes (\mathbb{S}/4)$.

Proof. The unit $\mathbb{S} \rightarrow ko$, where ko is the connective cover of KO , is an isomorphism in degrees zero, one, and two, carrying η and η^2 to the corresponding real K-theory classes; also $\pi_3 ko = 0$. Thus $H = \text{cofib}(\mathbb{S} \rightarrow ko)$ is four-connective, as is $H \otimes (\mathbb{S}/4)$. Cellular induction on Z gives

$$[Z, H \otimes (\mathbb{S}/4)] = [Z, \Sigma^{-1}(H \otimes (\mathbb{S}/4))] = 0.$$

The cofiber sequence therefore identifies $[Z, \mathbb{S}/4]$ with $[Z, ko \otimes (\mathbb{S}/4)]$. The latter target is the connective cover of $KO/4$, by the cofiber sequences for multiplication by four and $\pi_{-1} KO = 0$. Since Z is connective this gives the desired isomorphism. The two specified spectra have cells in degrees 0, 1 and 0, 1, 1, 2, respectively. $\square$

Theorem A.4. *For every $n \geq 3$, $A_n \otimes (\mathbb{S}/4)$ admits no unital homotopy-commutative multiplication with its specified unit. The same holds for $\mathrm{KO}/4$ with its quotient unit. In particular these pointed spectra admit no such $\mathbb{E}_2$ -ring structures.*

Proof. Write $i: \mathbb{S} \rightarrow \mathbb{S}/4$ and $\partial: \mathbb{S}/4 \rightarrow \Sigma\mathbb{S}$ for the Moore cofiber maps, and let T interchange the two factors of $(\mathbb{S}/4)^{\otimes 2}$. Oka’s calculation says that every unital product λ on $\mathbb{S}/4$ satisfies

$$(A.1) \quad \lambda T - \lambda = i\eta^2(\partial \otimes \partial) \neq 0,$$

where the right side has order two [Oka84, Theorem 2(c),(c’)].

Suppose a pointed spectrum (B, v) has maps $j: \mathbb{S}/4 \rightarrow B$ and $r: B \rightarrow \mathrm{KO}/4$ with $ji = v$ and $rj = e$. A unital homotopy-commutative product μ on B would give $r\mu(j \otimes j): (\mathbb{S}/4)^{\otimes 2} \rightarrow \mathrm{KO}/4$. Lemma A.3 lifts its homotopy class uniquely to a product λ on $\mathbb{S}/4$. Both unit equations follow by injectivity on $[\mathbb{S}/4, -]$, and commutativity follows by injectivity on $[(\mathbb{S}/4)^{\otimes 2}, -]$. This contradicts (A.1).

For $B = \mathrm{KO}/4$, take $j = e$ and $r = \mathrm{id}$. For $B = A_n \otimes (\mathbb{S}/4)$ with $n \geq 3$, the cofiber model of Proposition 2.3 and $\eta^3 = 0$ give a KO -module retraction $\rho_n: A_n \rightarrow \mathrm{KO}$ of the unit. Take j induced by the unit of A_n and $r = \rho_n \otimes \mathrm{id}_{\mathbb{S}/4}$. These satisfy the required identities; no multiplicativity of j or r is needed. $\square$

Theorems A.2 and A.4 are obstructions to specified multiplications, not assertions that a Picard group vanishes. The mod-four argument requires a retraction onto KO , absent at the first two eta stages. These constructions and obstructions do not settle intrinsic $\mathbb{E}_2$ structures for $m = 2$, $n = 1, 2$, or for $m = 3, 4$, $n \geq 2$. No numerical Picard group for these unspecified algebras is inferred from Theorem 1.3.

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